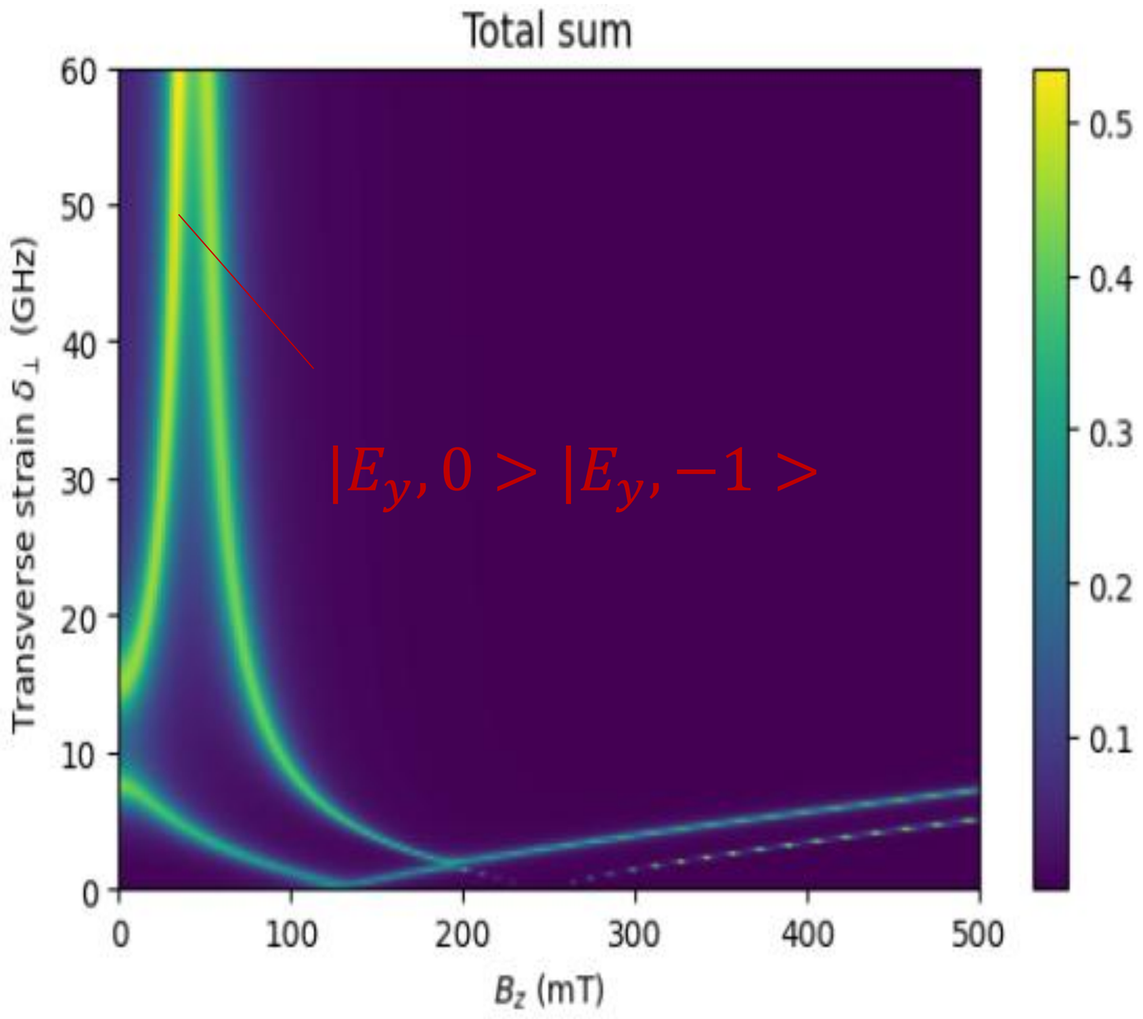
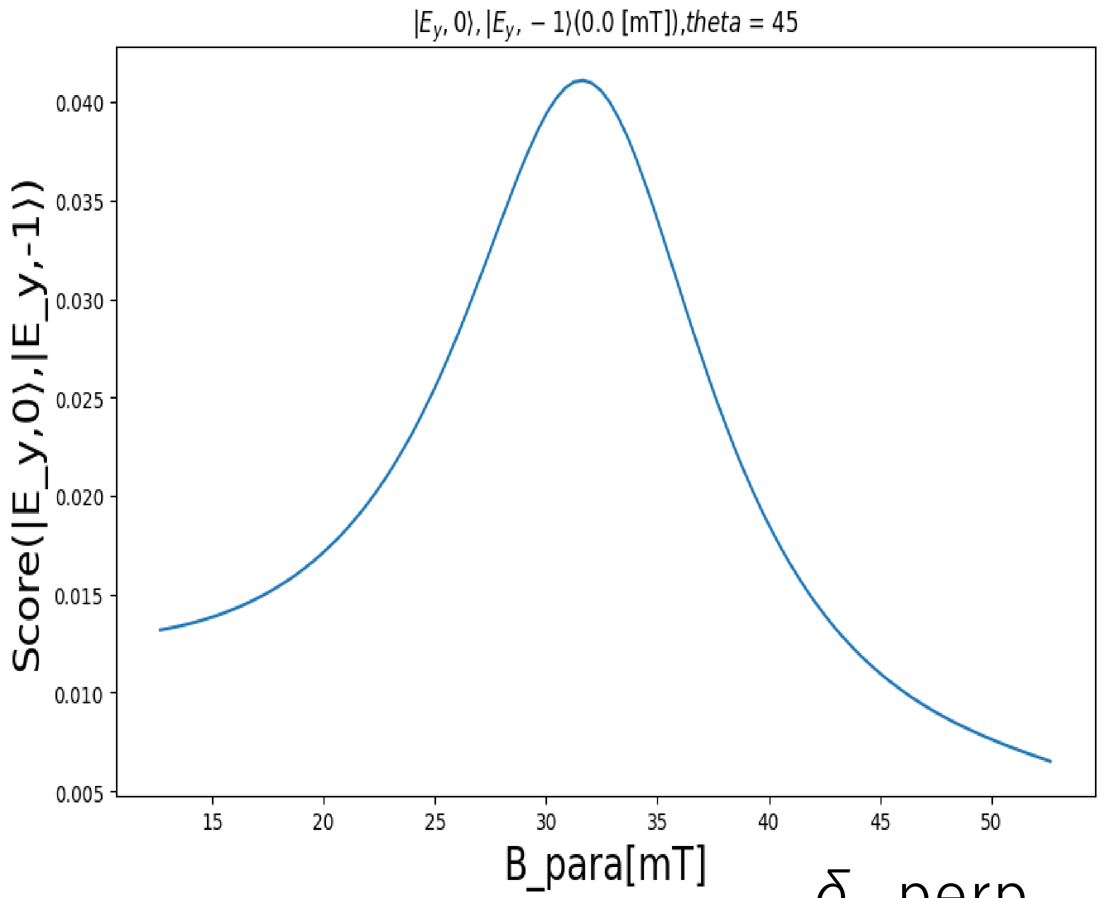
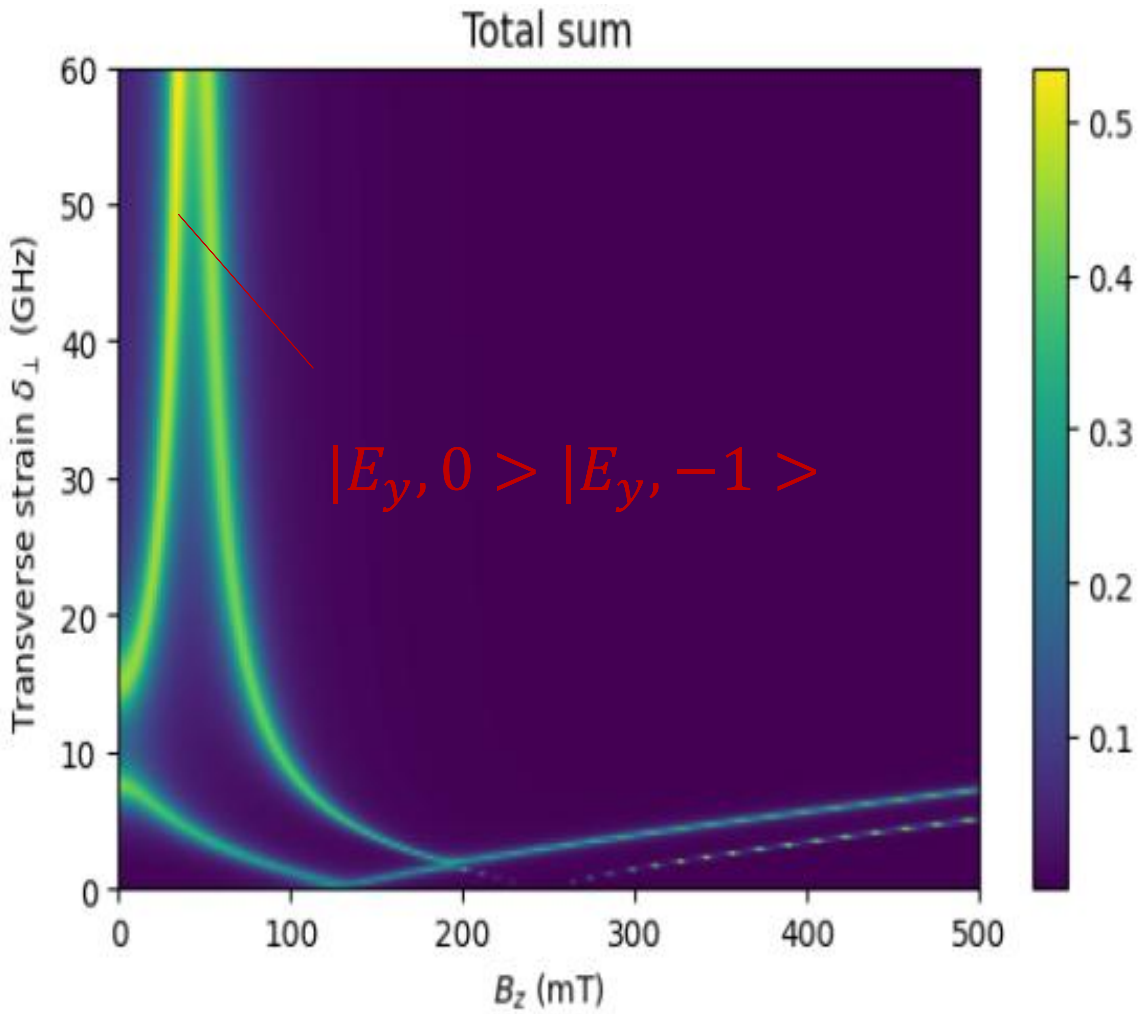
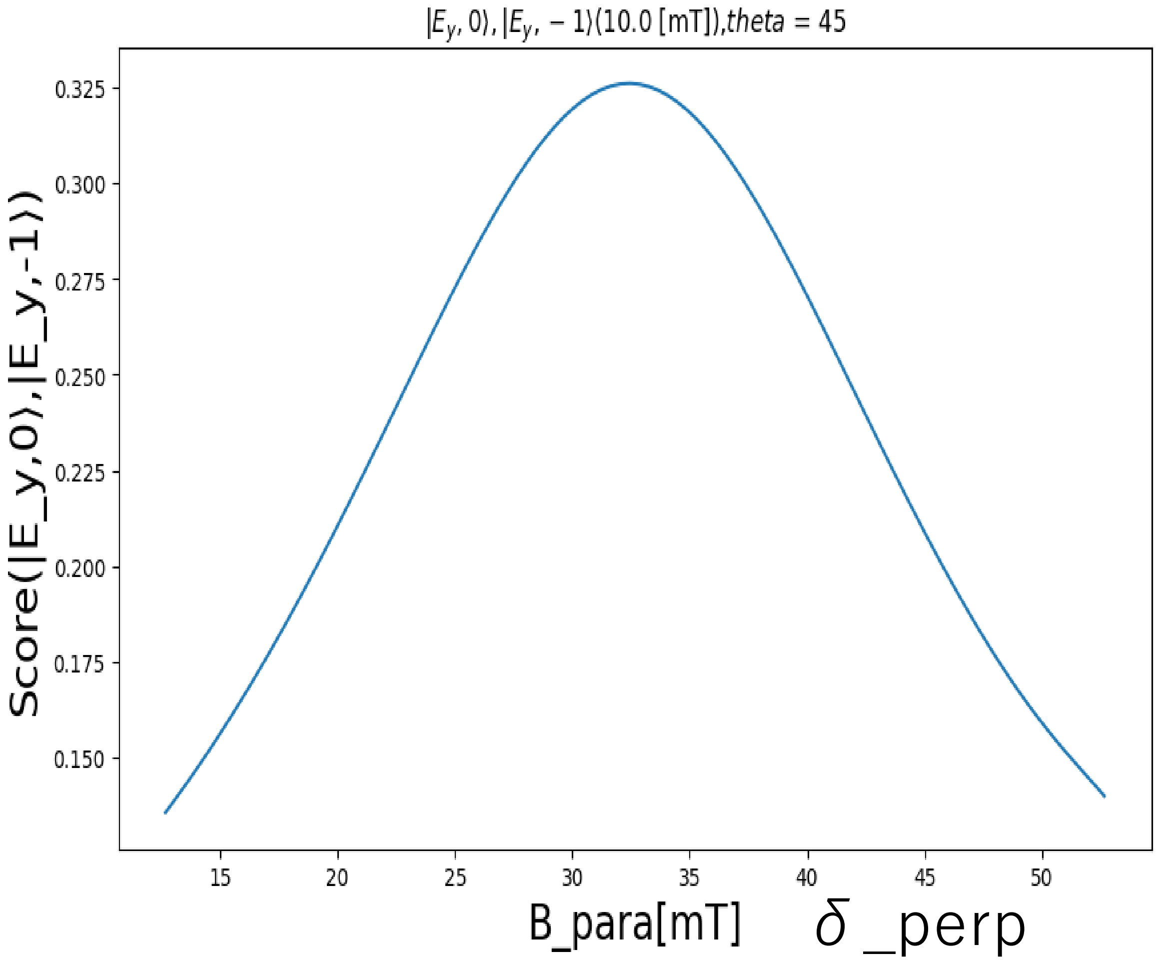


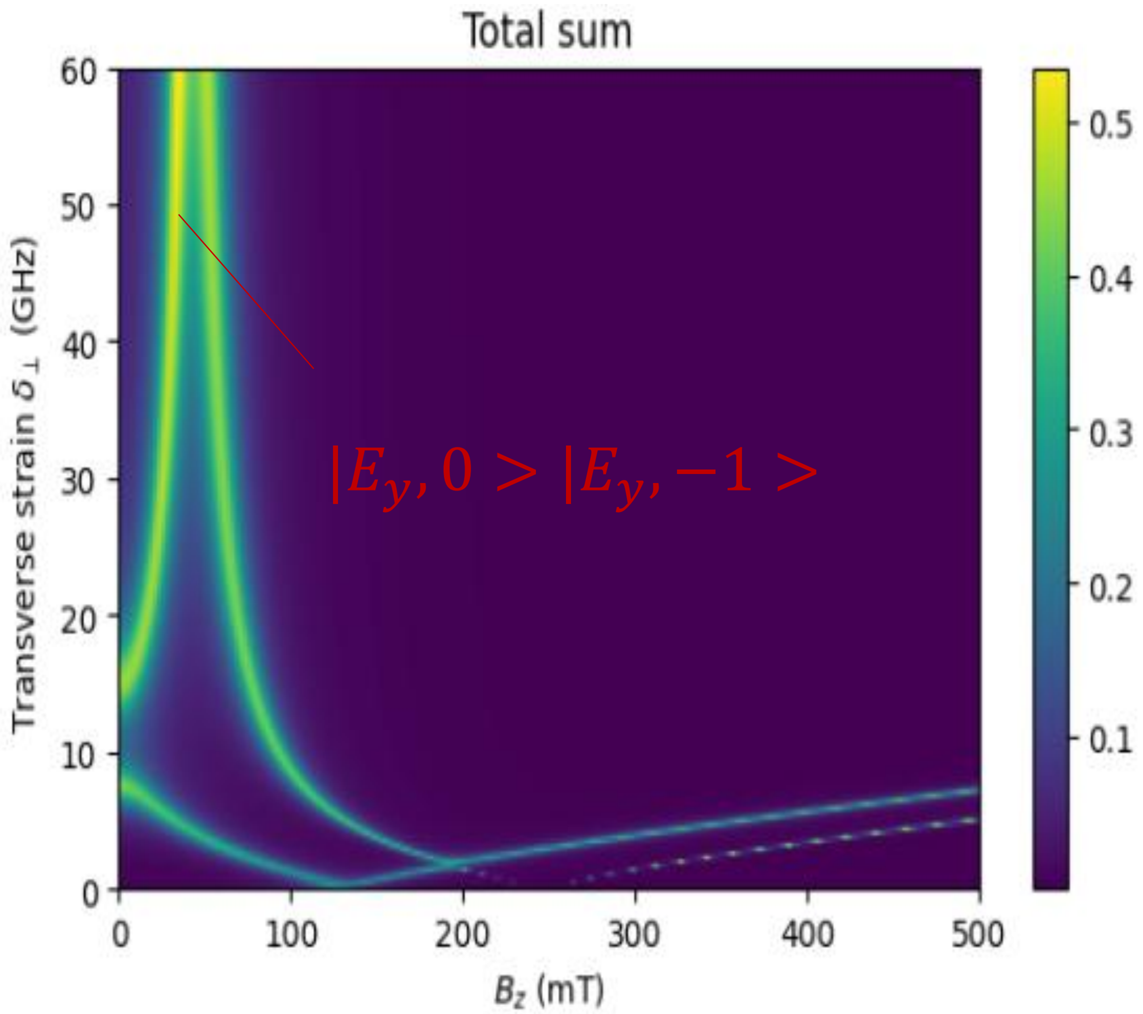


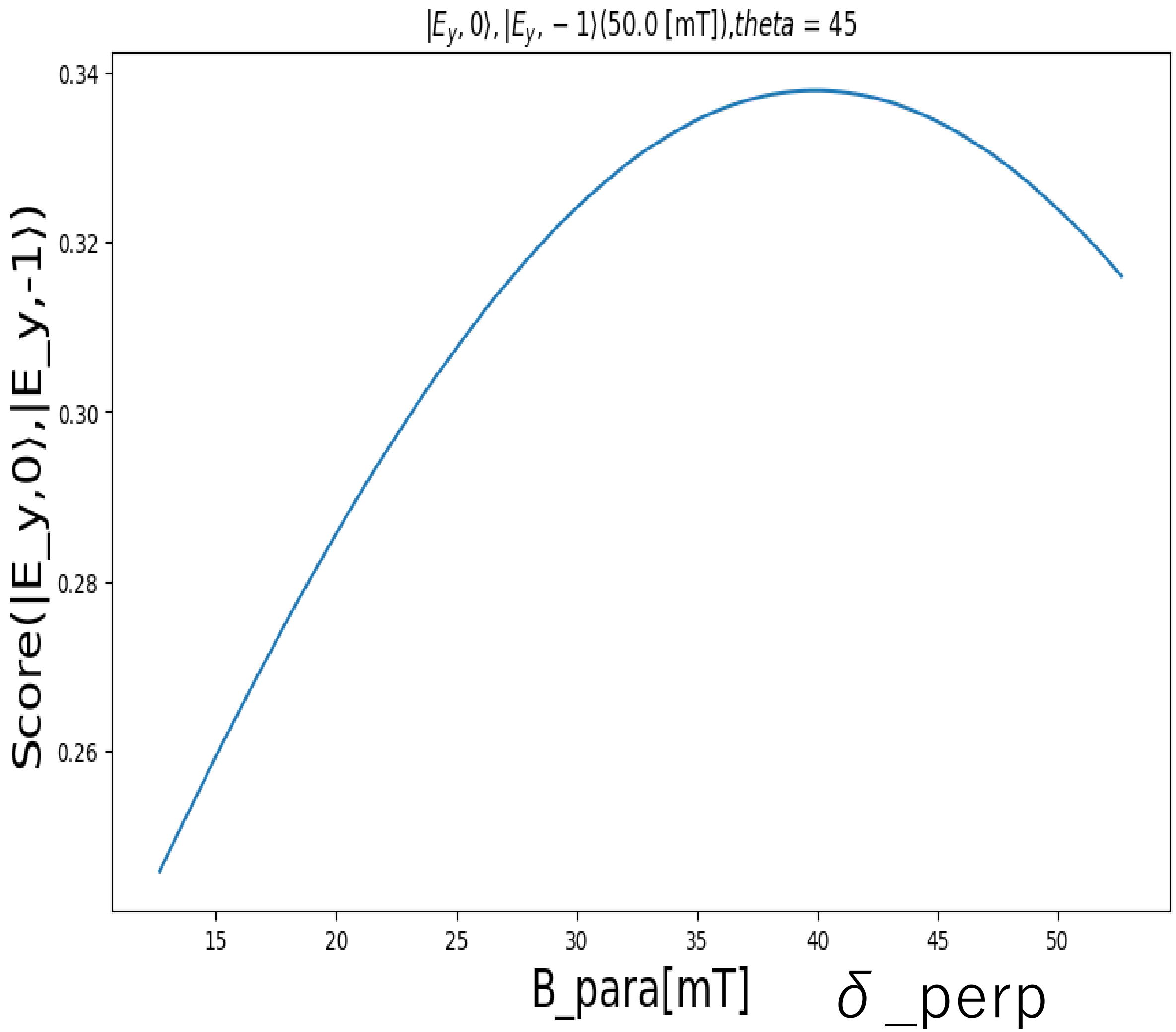
δ _perp
= 50.0GHZ



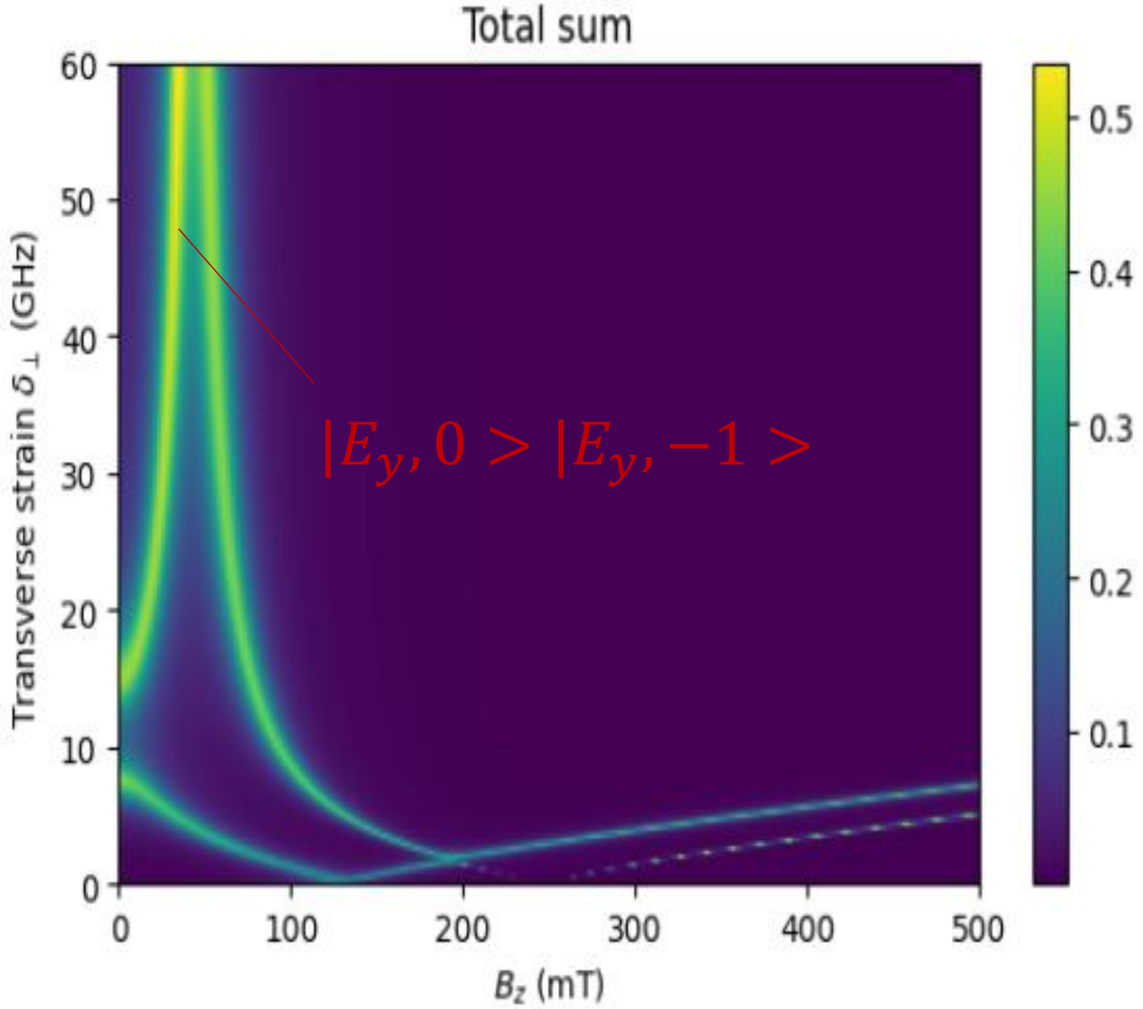


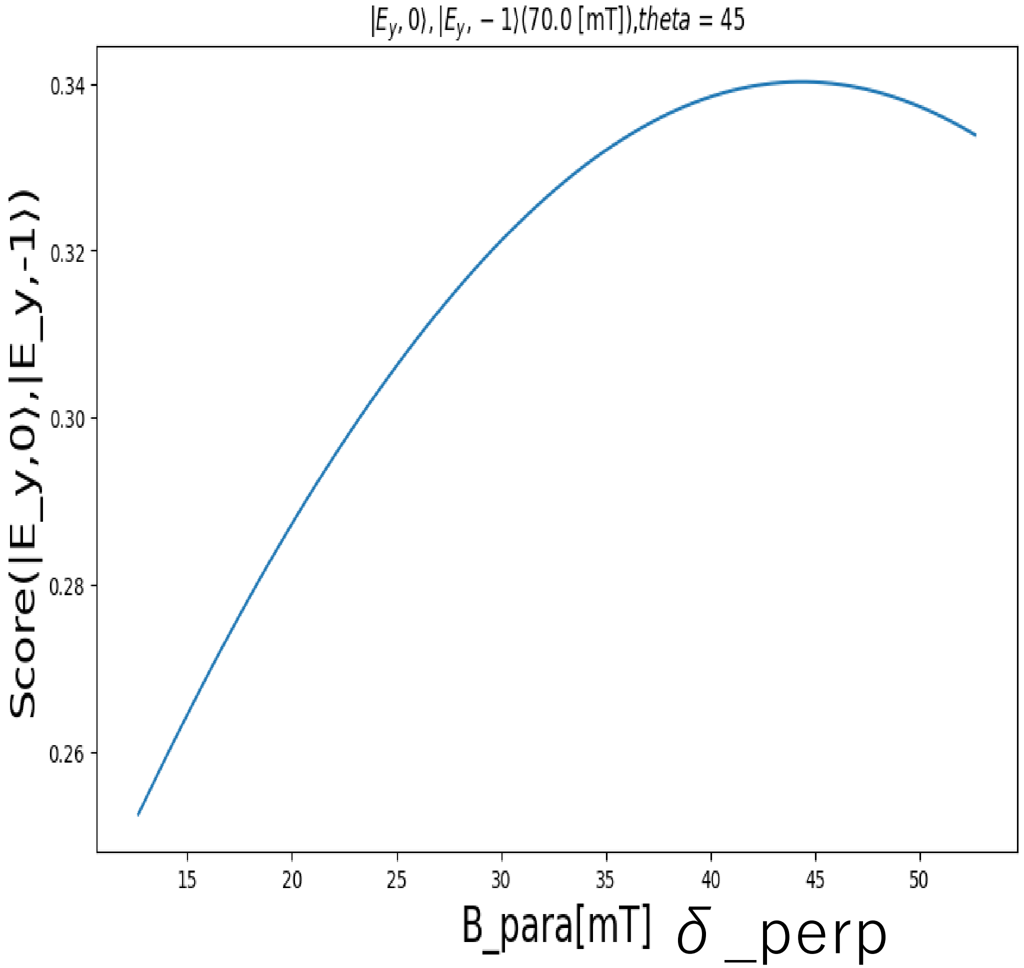
δ _perp = 50.0GHZ



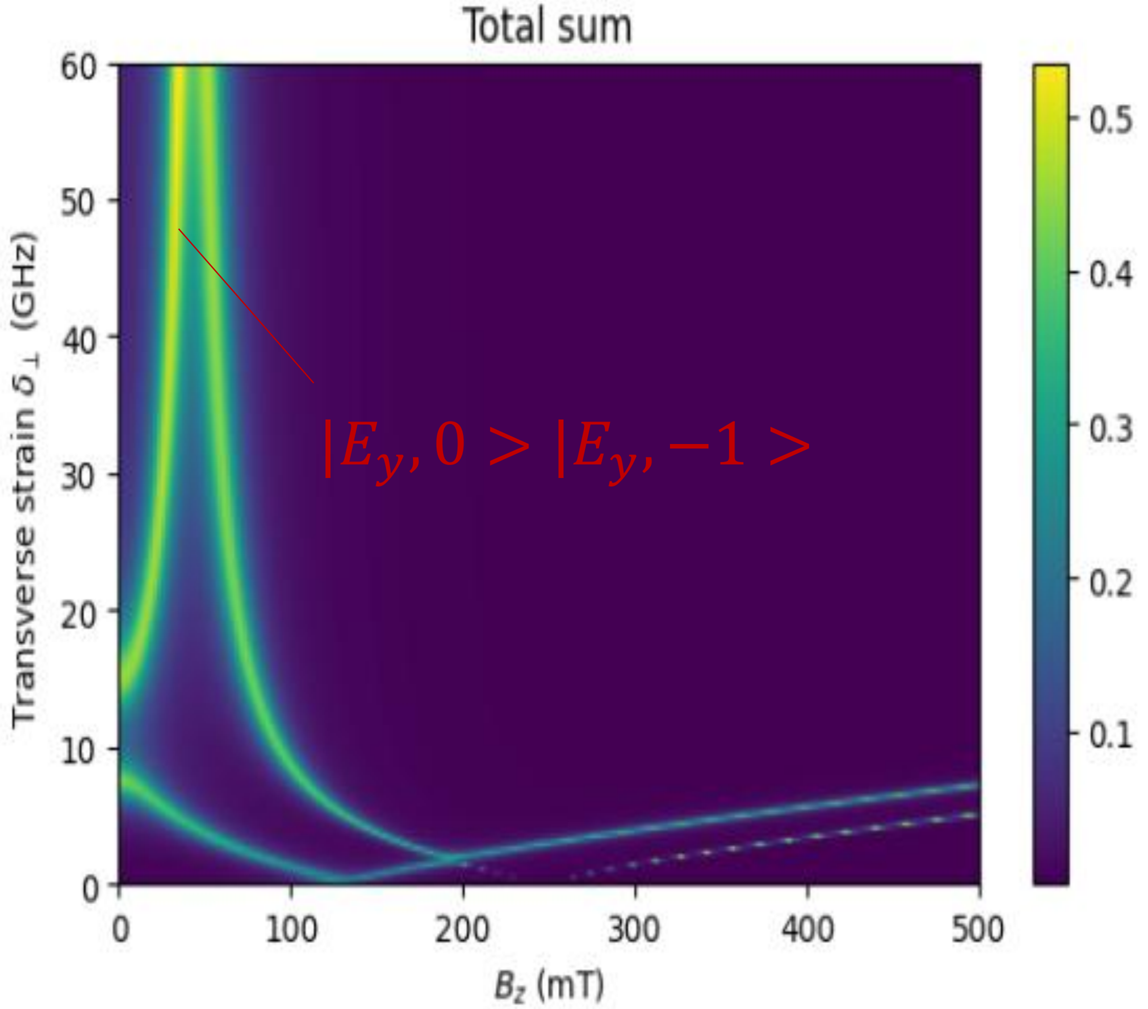


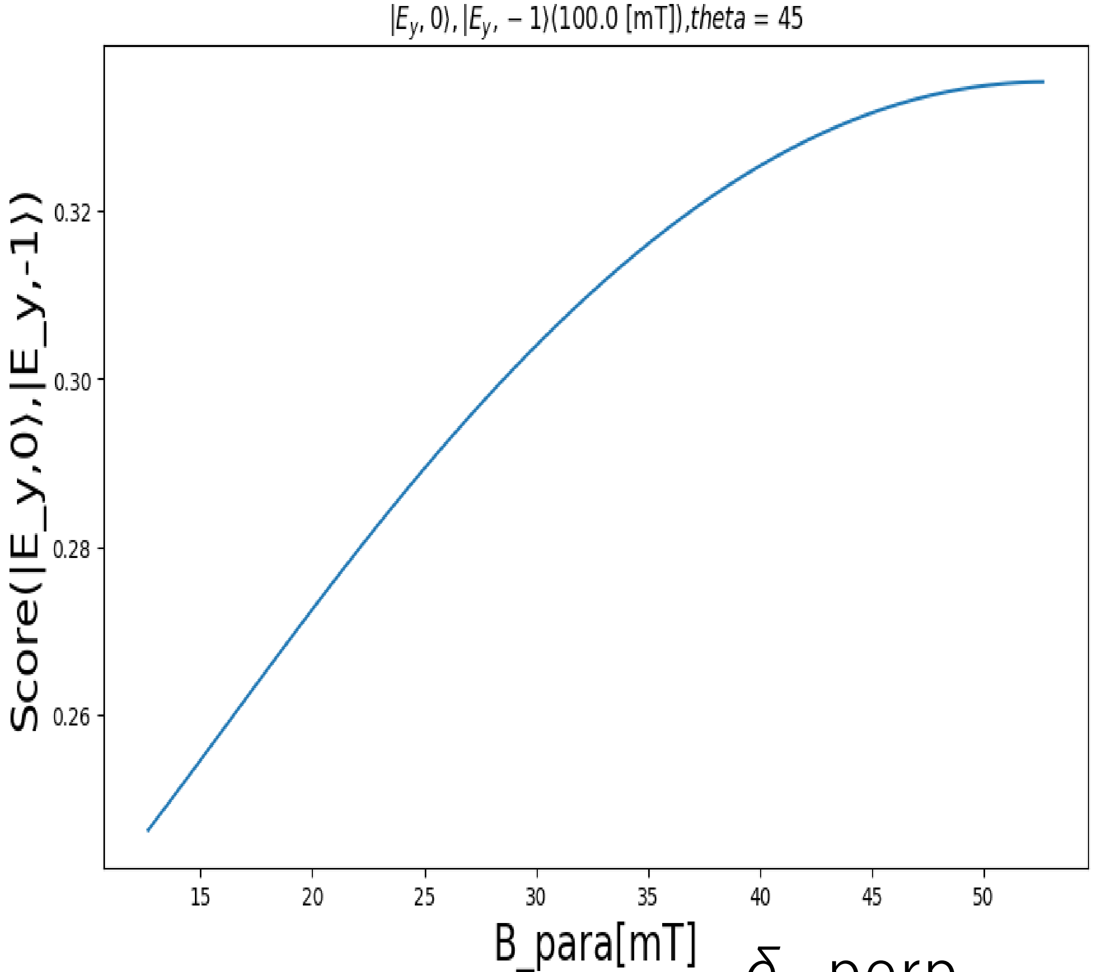
δ_perp = 50.0GHZ





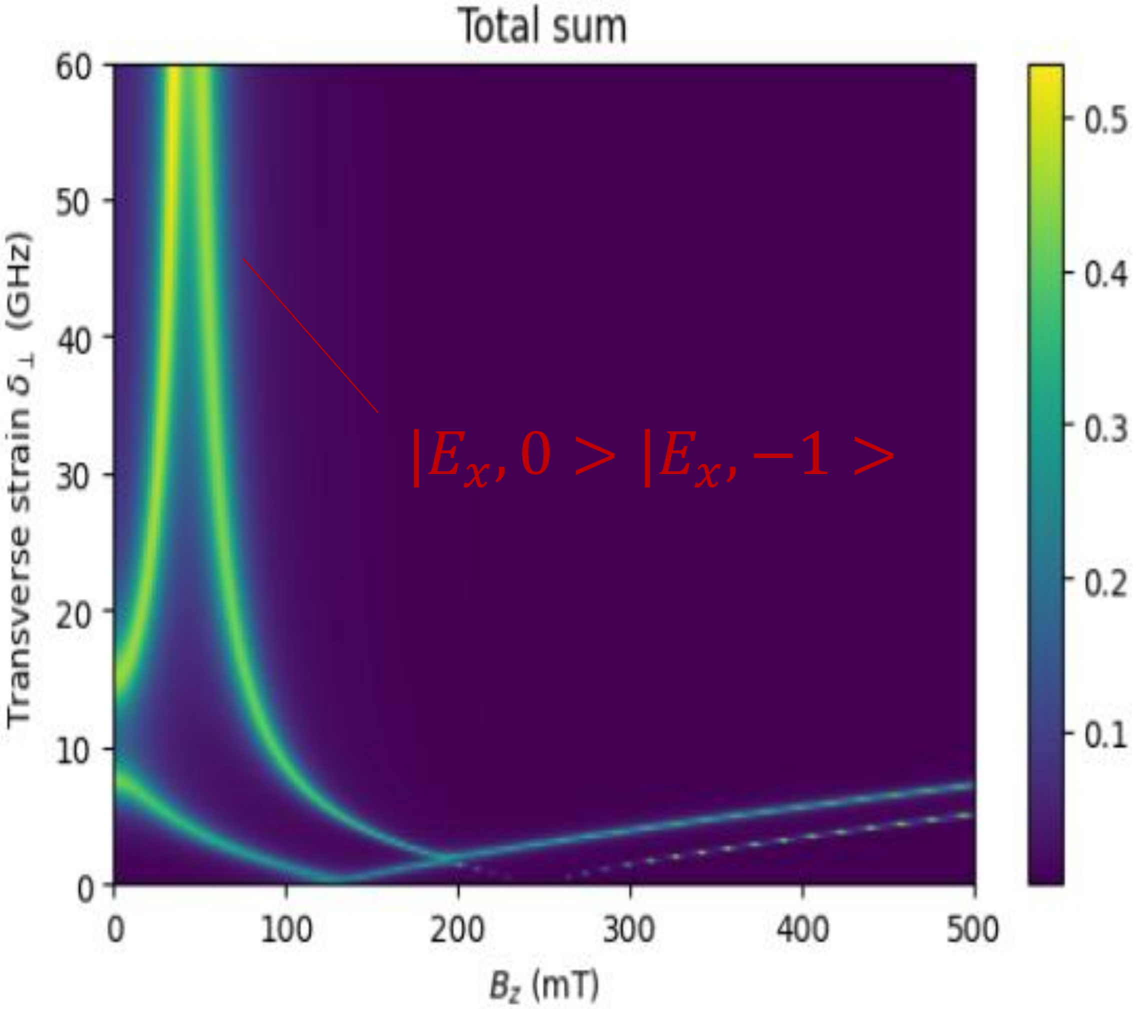
δ_perp = 50.0GHZ

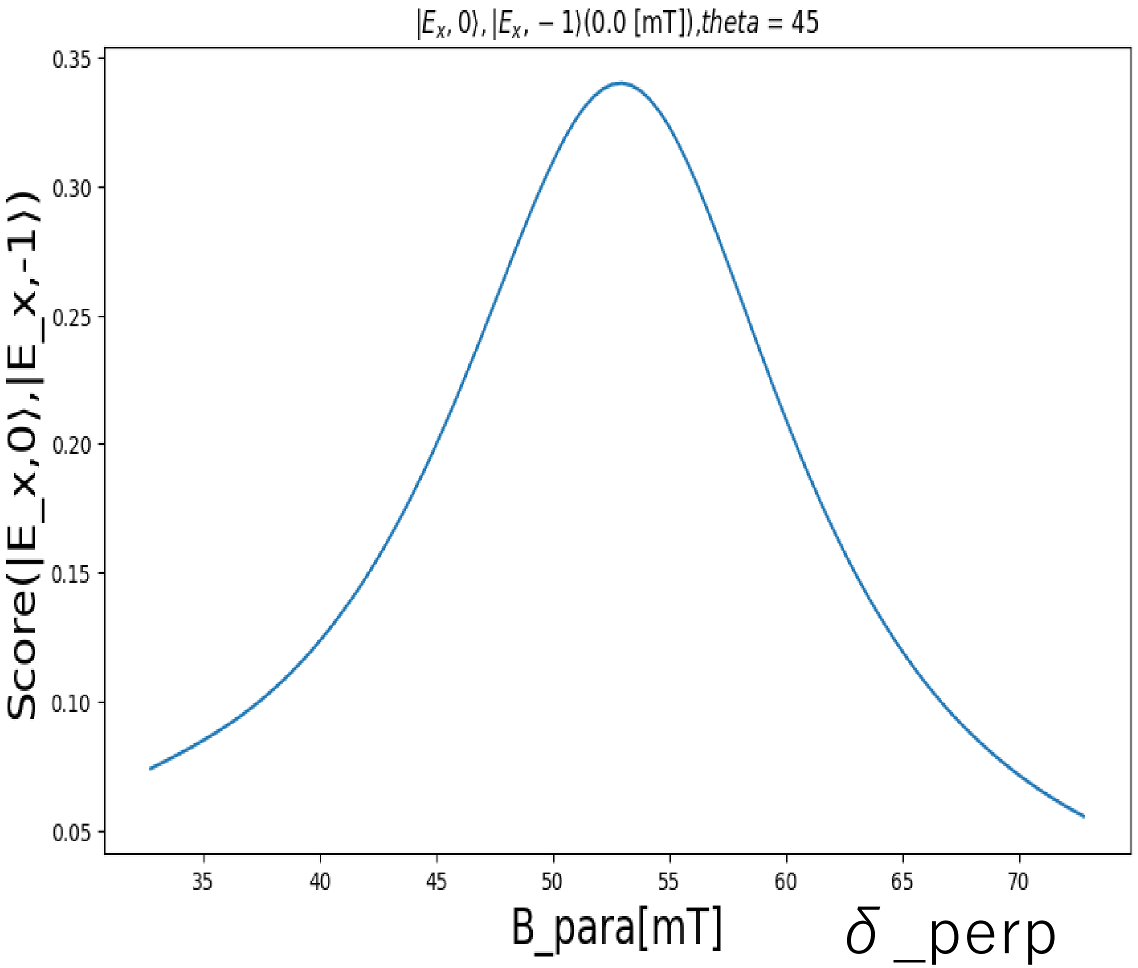




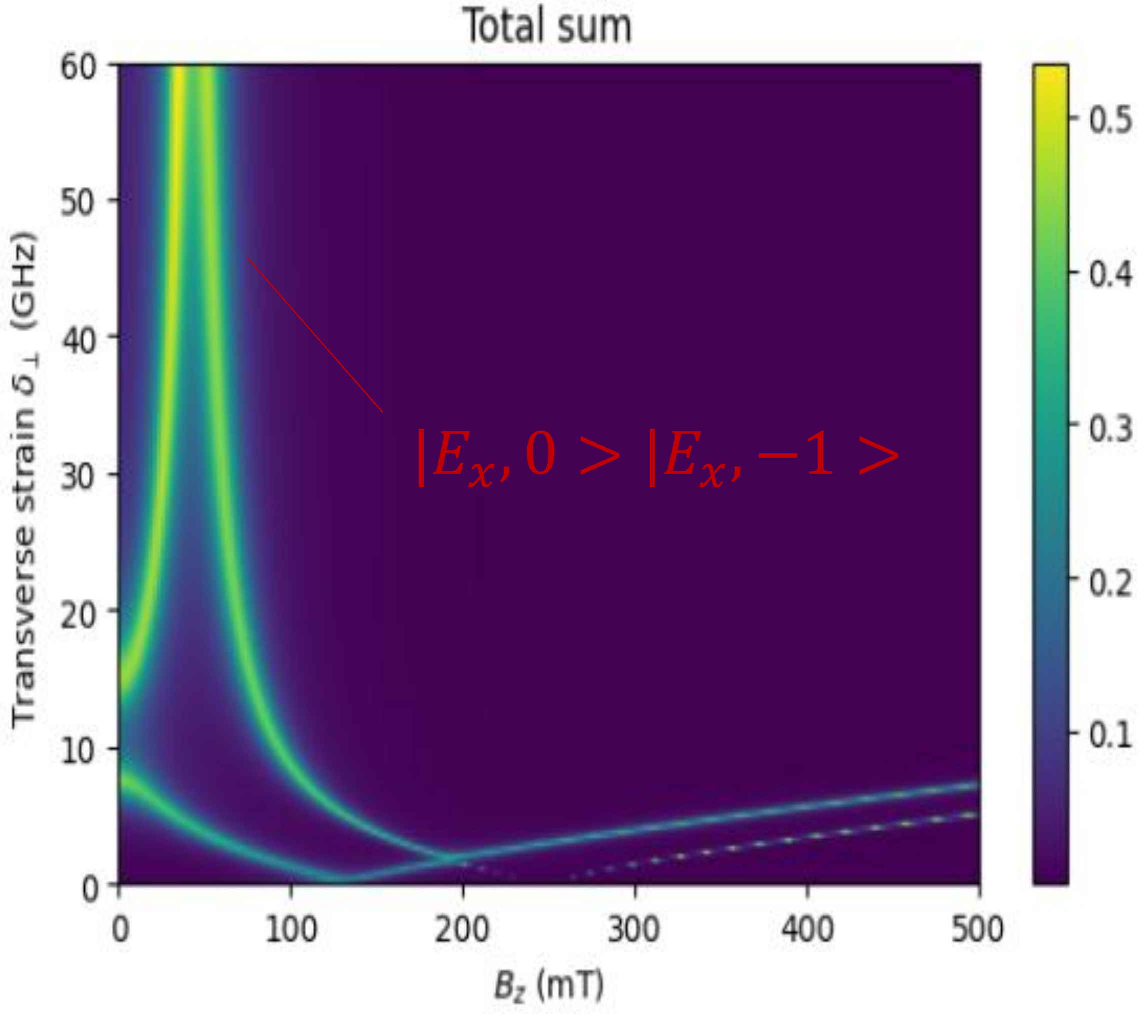
δ_perp
= 50.0GHZ

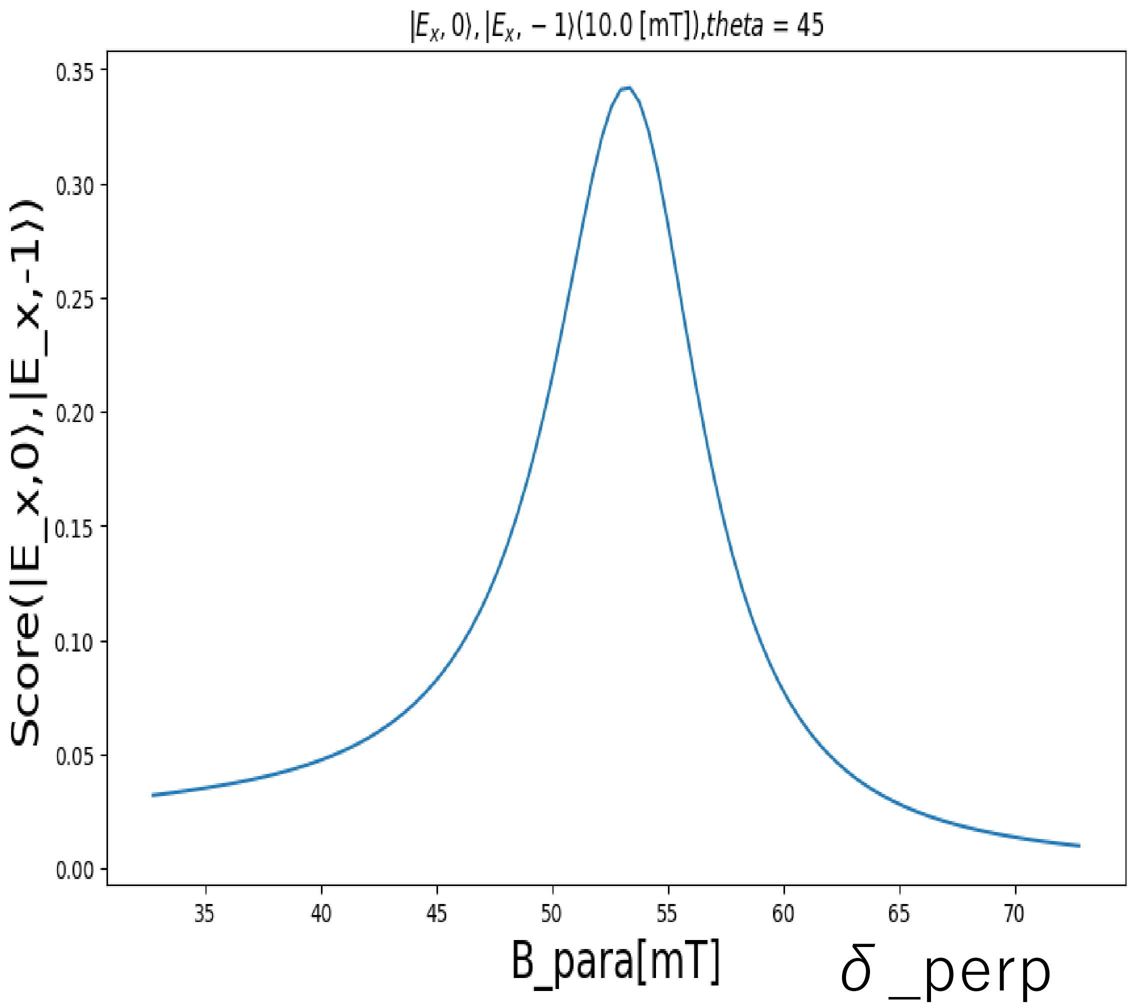
・混ざり具合($E_y, 0 > |E_y, -1 >$)の線幅が広くなる
・その最大値となる点が右にシフト



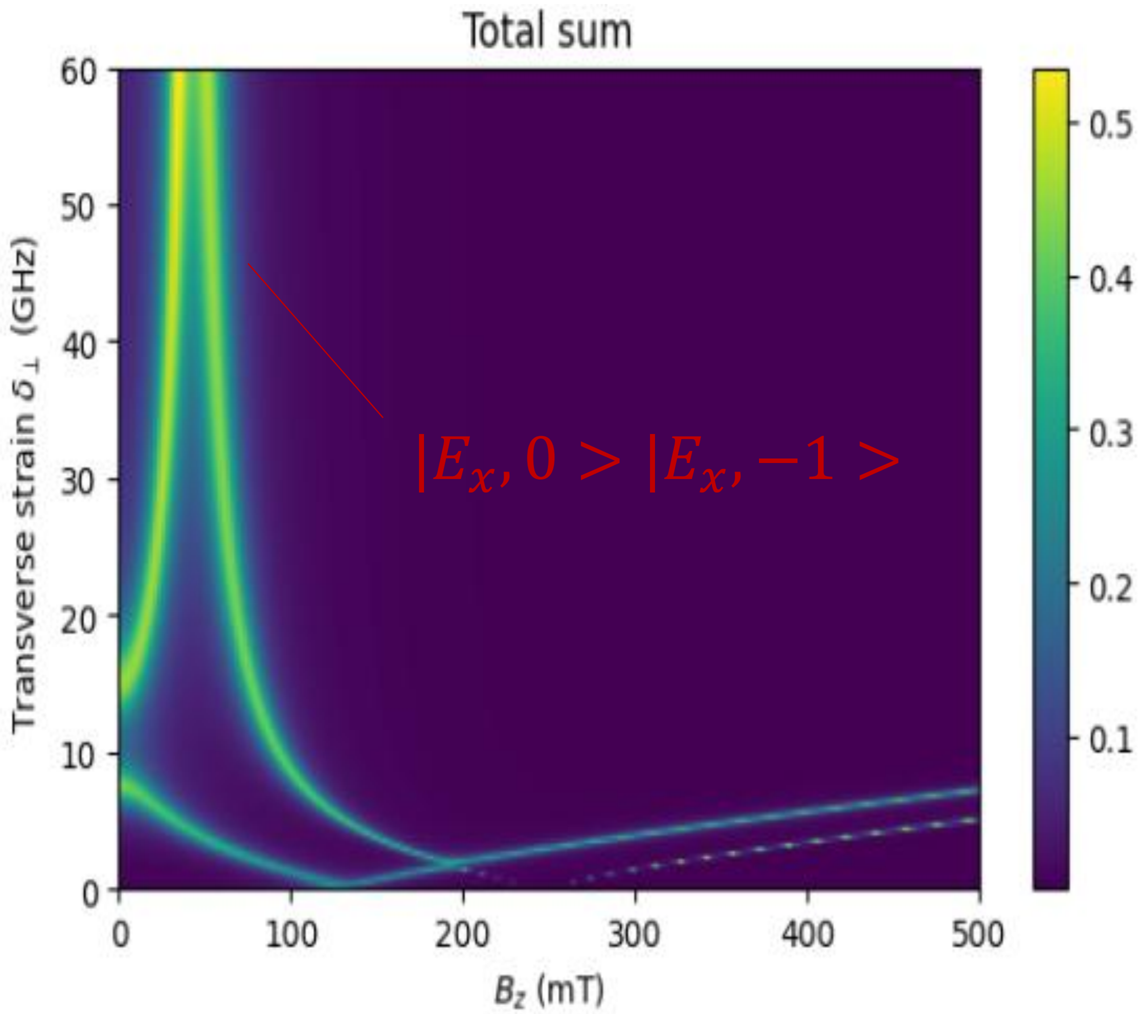


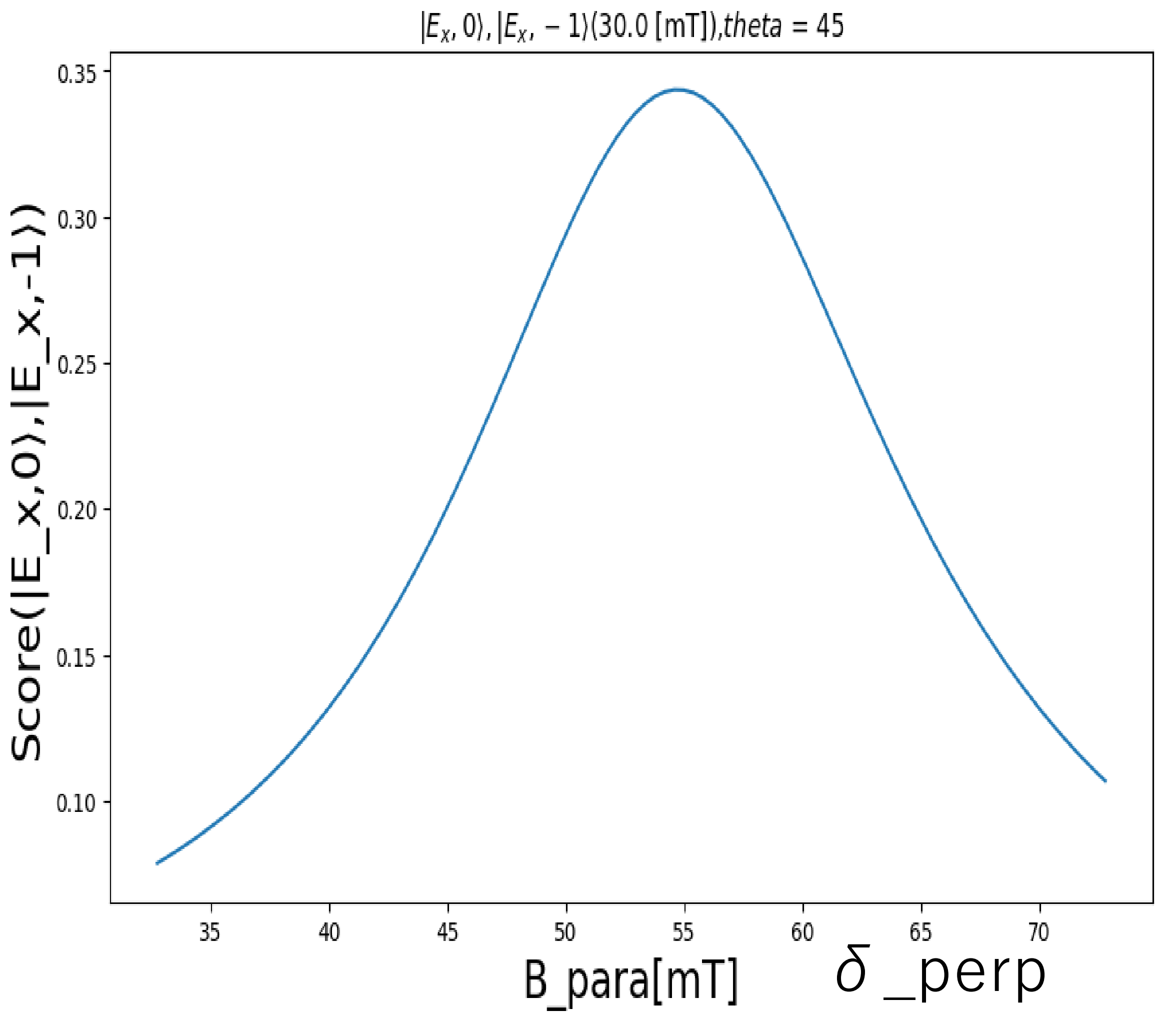
δ_perp
= 50.0GHZ



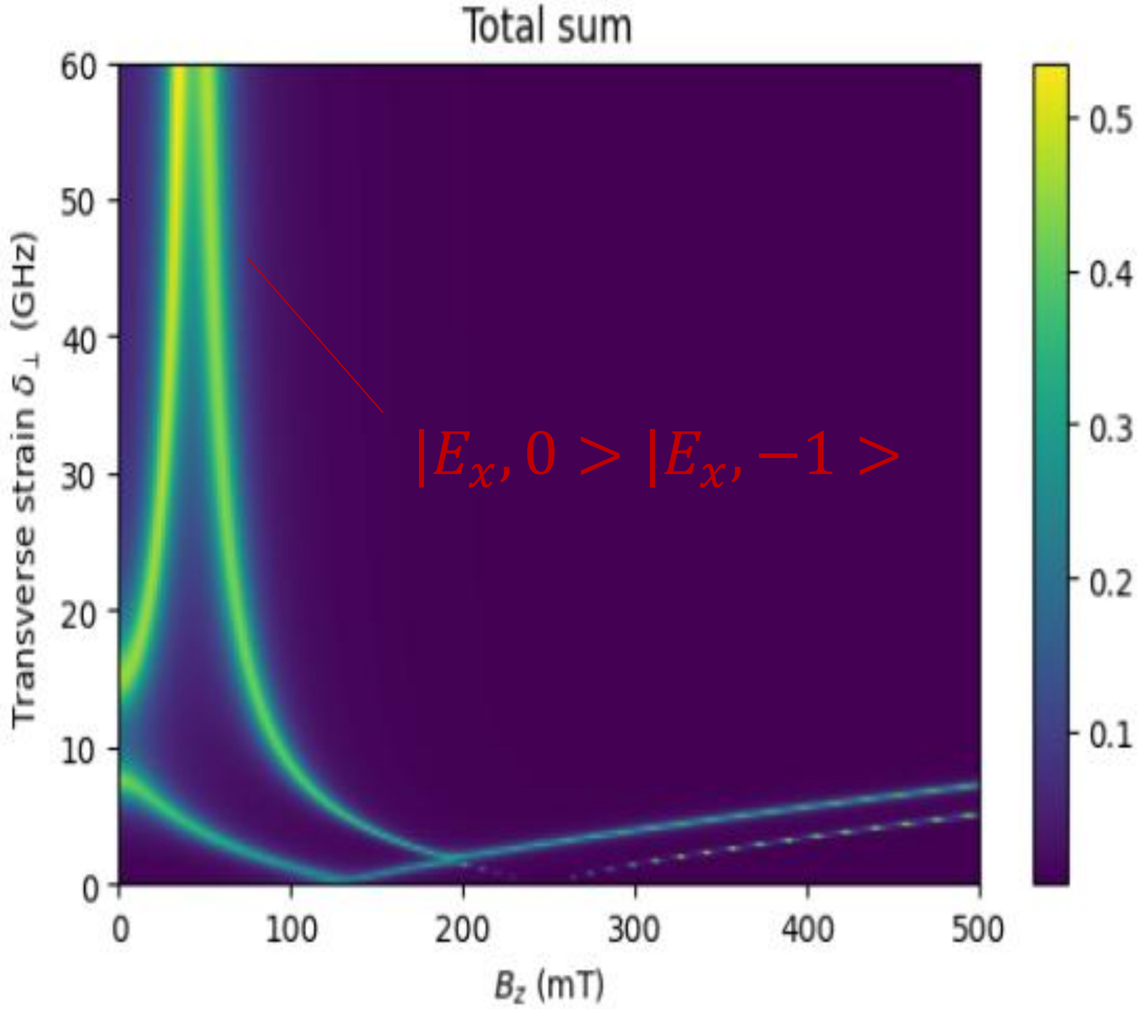


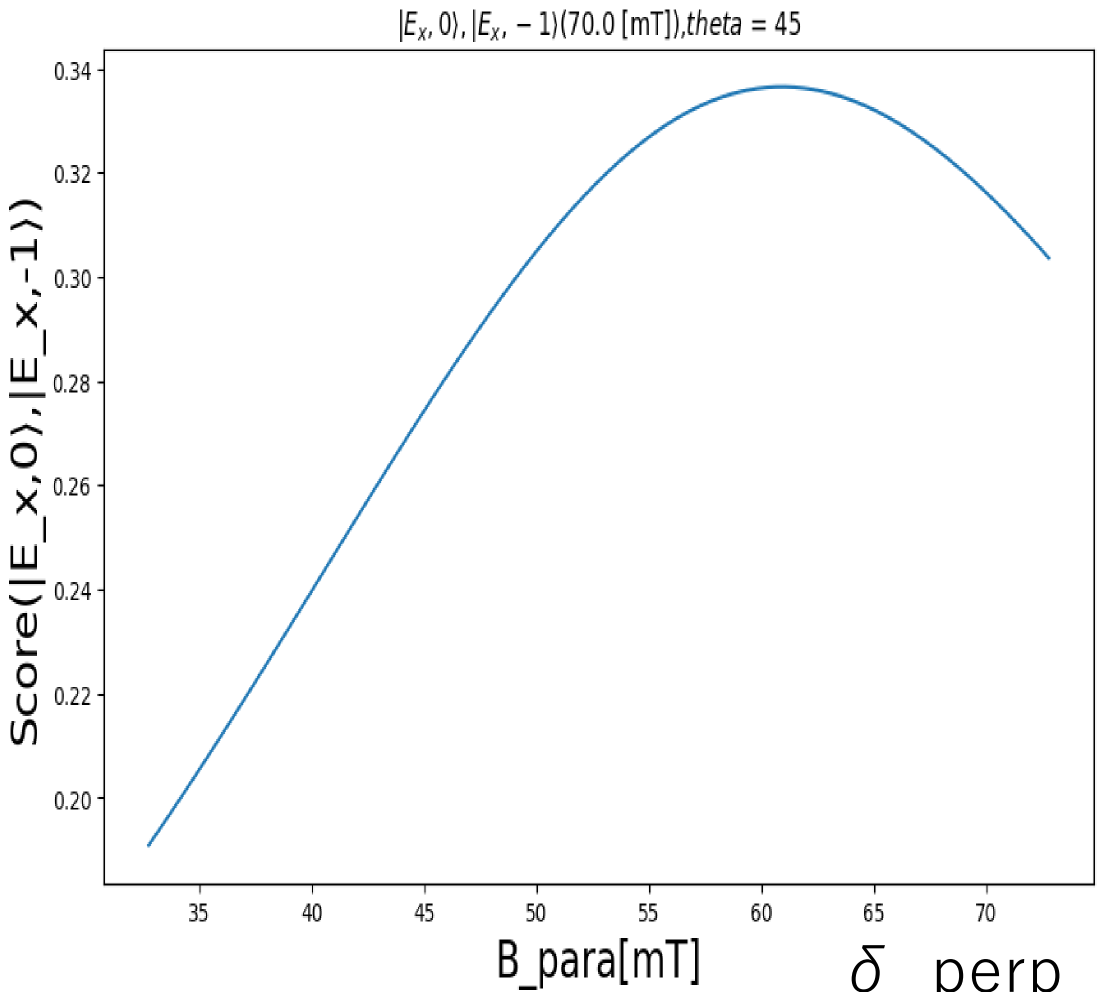
δ _perp
＝ 50.0GHZ



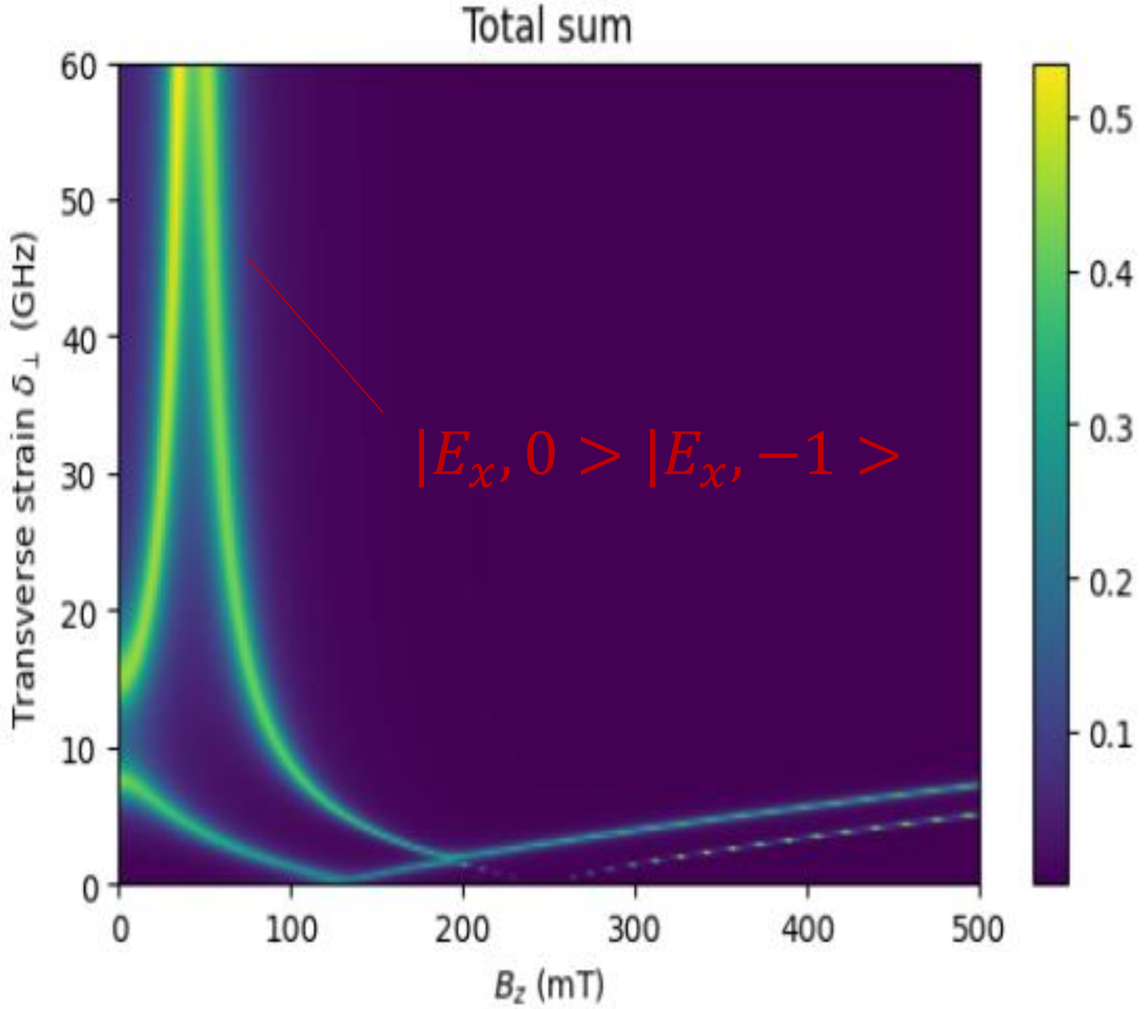


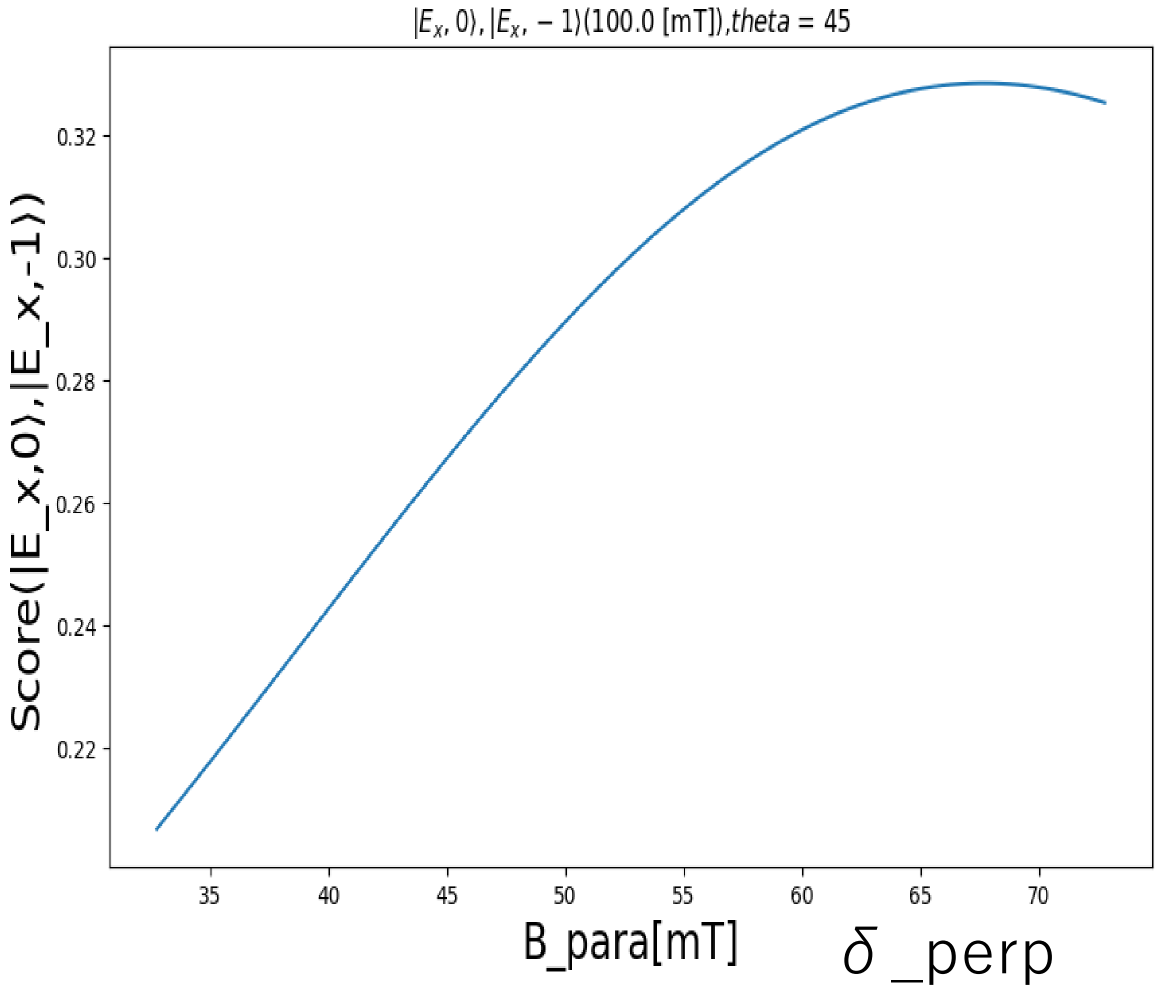
δ _perp
= 50.0GHZ





δ _perp
= 50.0GHZ

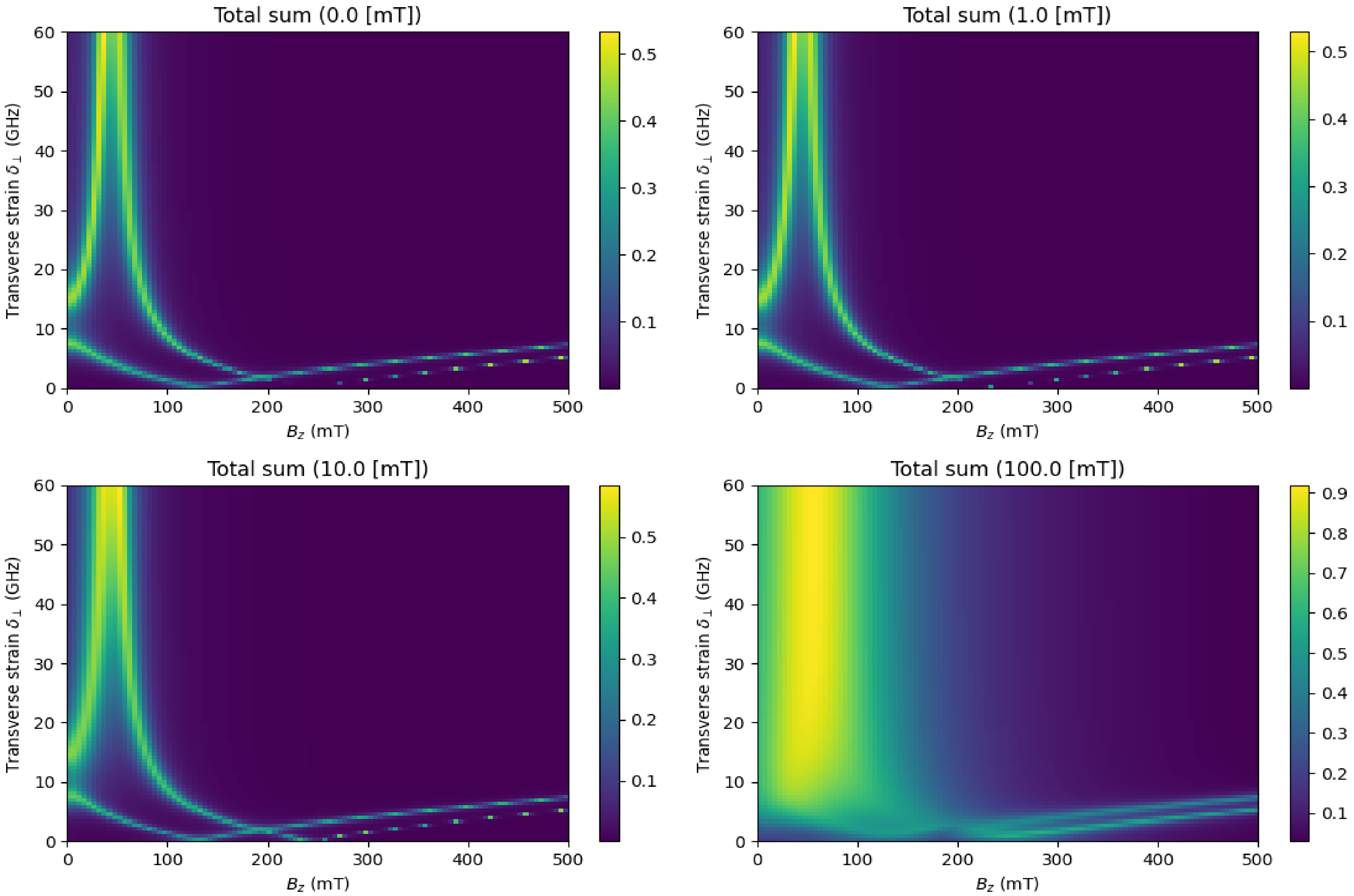




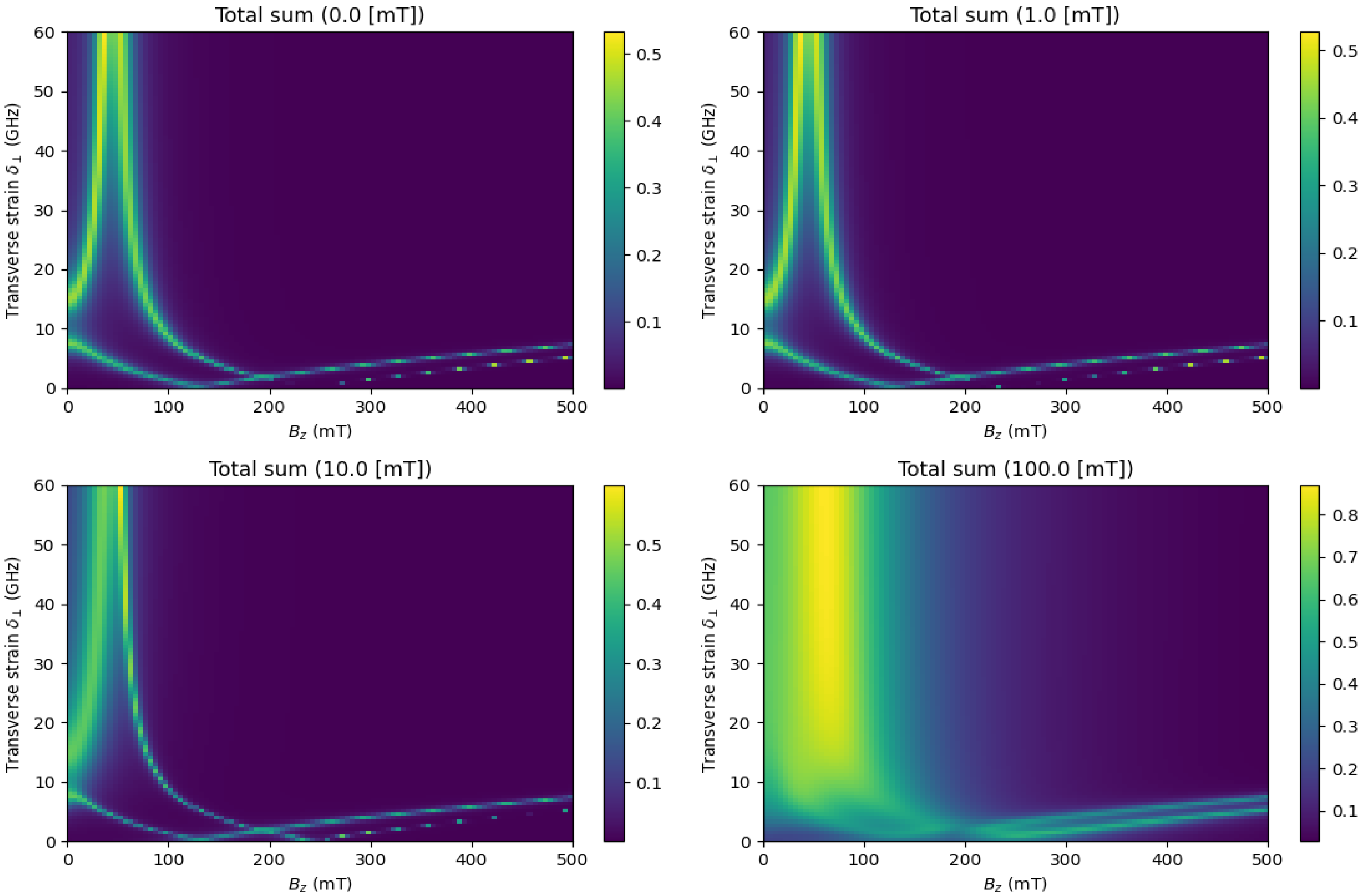
δ_perp
= 50.0GHZ

- 混ざり具合($E_x, 0 > |E_x, -1 >$)の線幅は10mTで一旦狭くなり、あとは広がる
- その最大値となる点が右にシフトするが、その移動は ($E_y, 0 > |E_y, -1 >$)に比べて穏やか

$\theta = 0°$

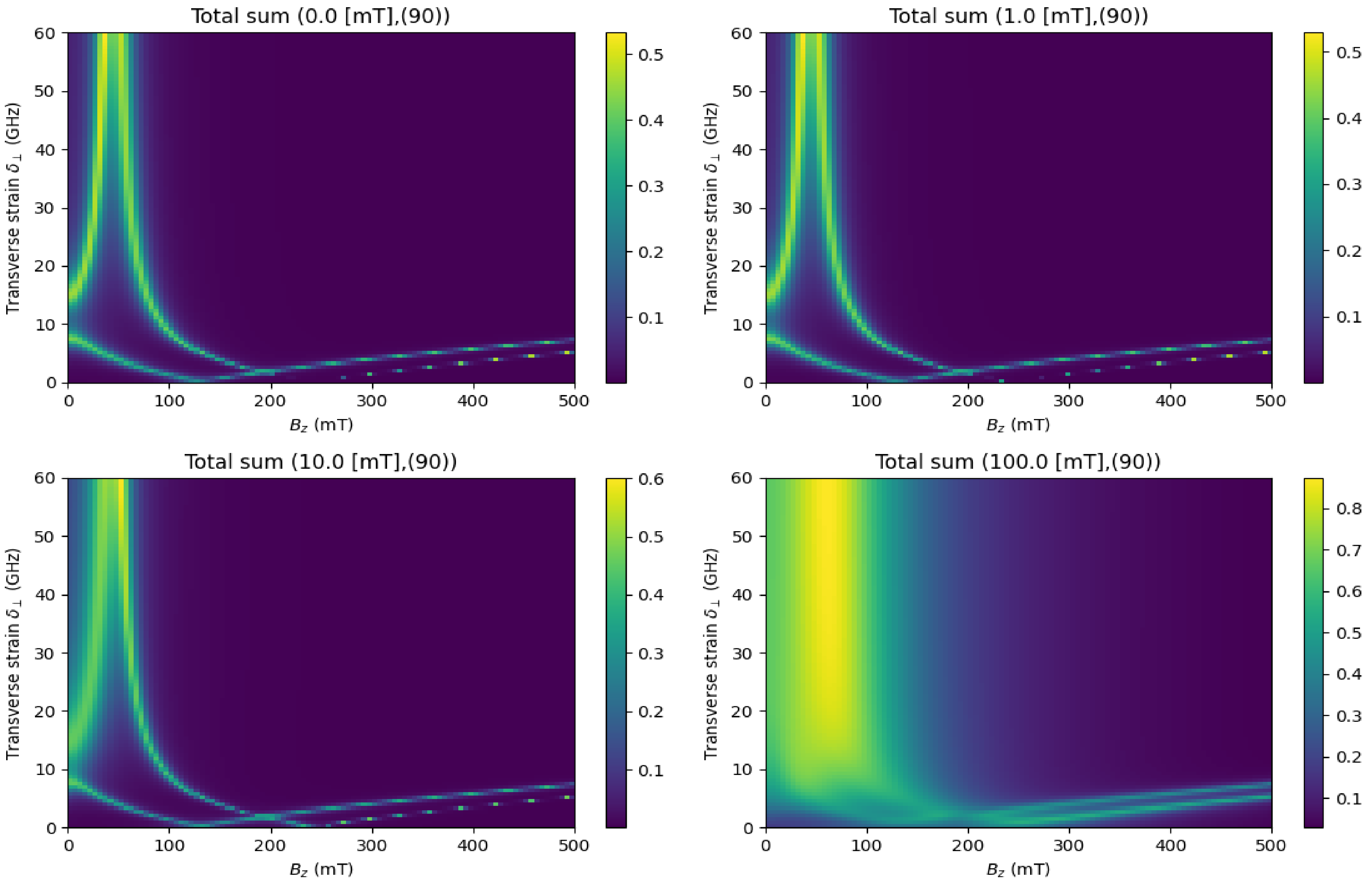

Total sum (0.0 [mT])
Total sum (1.0 [mT])
Total sum (10.0 [mT])
Total sum (100.0 [mT])
Transverse strain $\delta_\perp$ (GHz)
B_z (mT)
0
10
20
30
40
50
60
100
200
300
400
500
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9

$\theta = 45°$



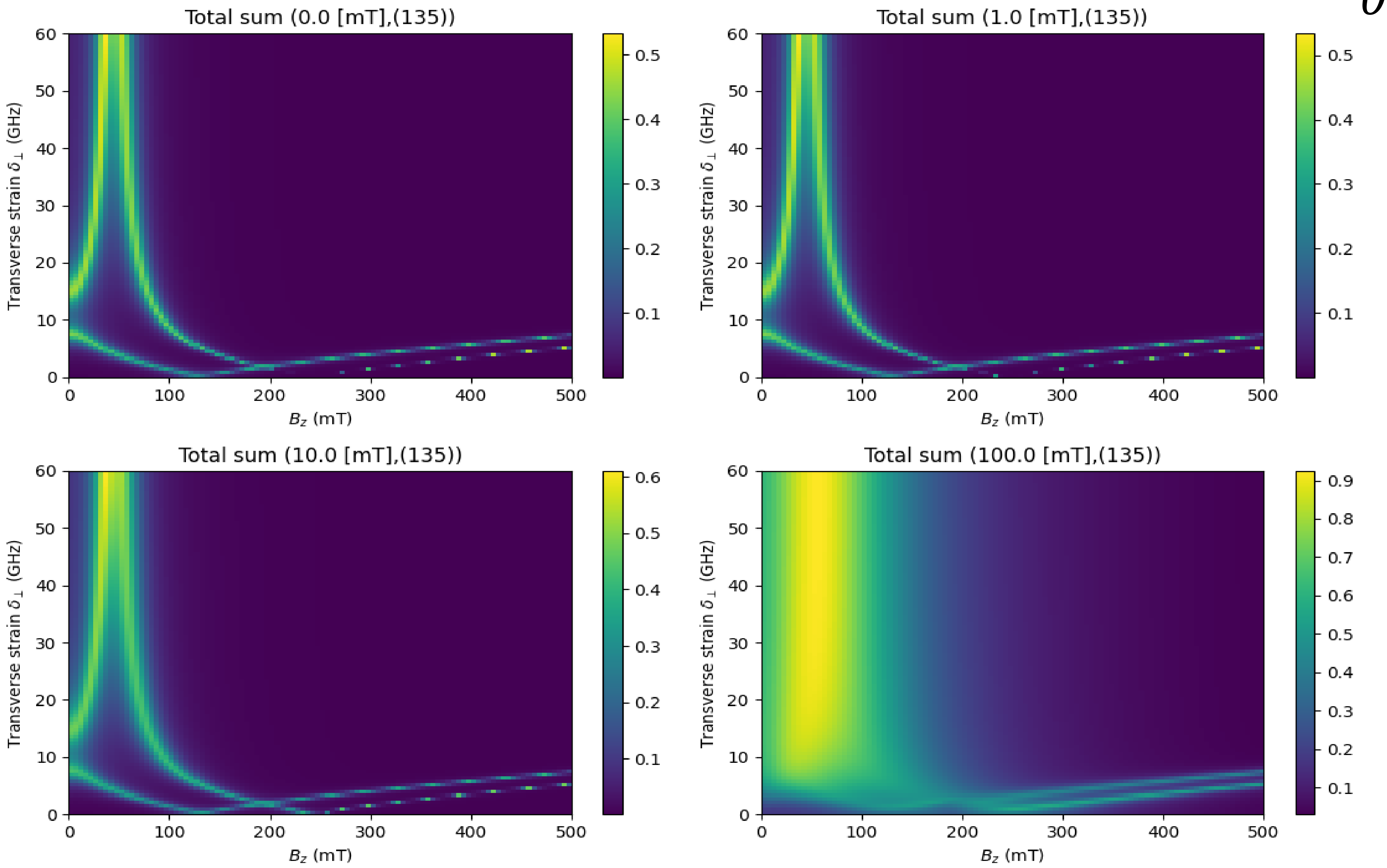
Total sum (0.0 [mT])
Total sum (1.0 [mT])
Total sum (10.0 [mT])
Total sum (100.0 [mT])
Transverse strain $\delta_\perp$ (GHz)
B_z (mT)
0
100
200
300
400
500
10
20
30
40
50
60
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8

$\theta = 90°$



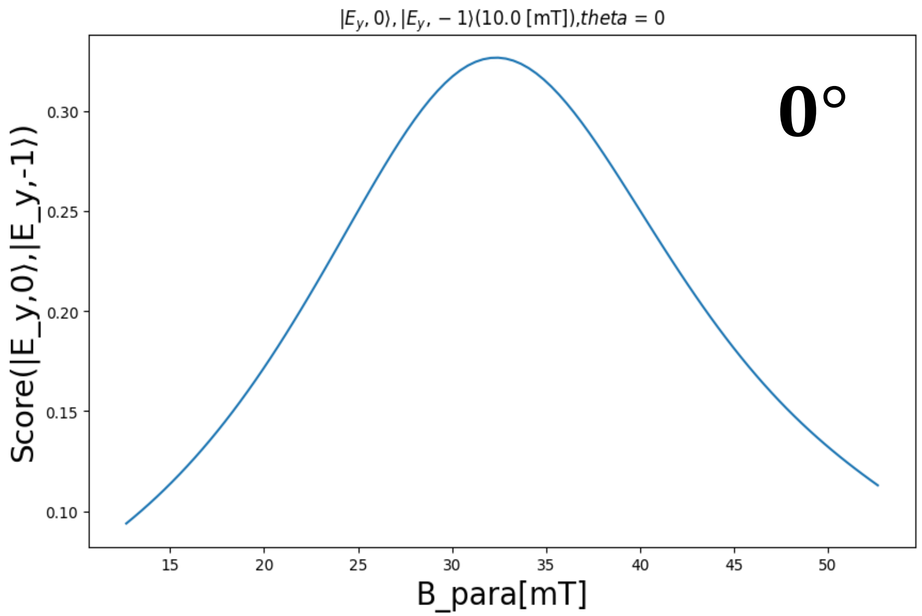
Total sum (0.0 [mT],(90))
Total sum (1.0 [mT],(90))
Total sum (10.0 [mT],(90))
Total sum (100.0 [mT],(90))
Transverse strain $\delta_\perp$ (GHz)
B_z (mT)

$\theta = 135°$

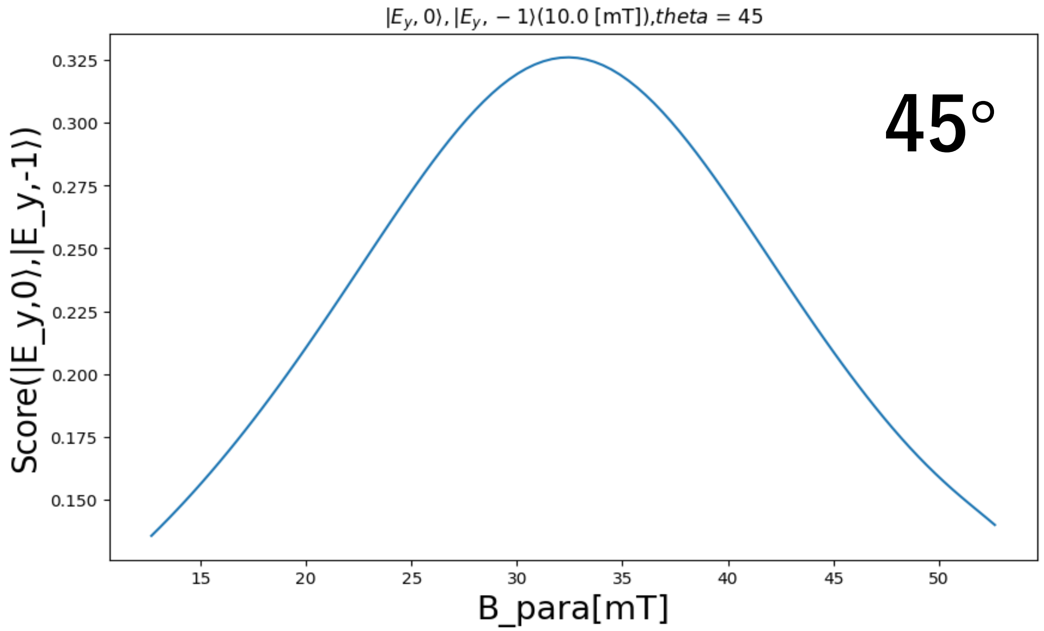


Total sum (0.0 [mT],(135))
Total sum (1.0 [mT],(135))
Total sum (10.0 [mT],(135))
Total sum (100.0 [mT],(135))
Transverse strain $\delta_\perp$ (GHz)
B_z (mT)
0
100
200
300
400
500
10
20
30
40
50
60
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9

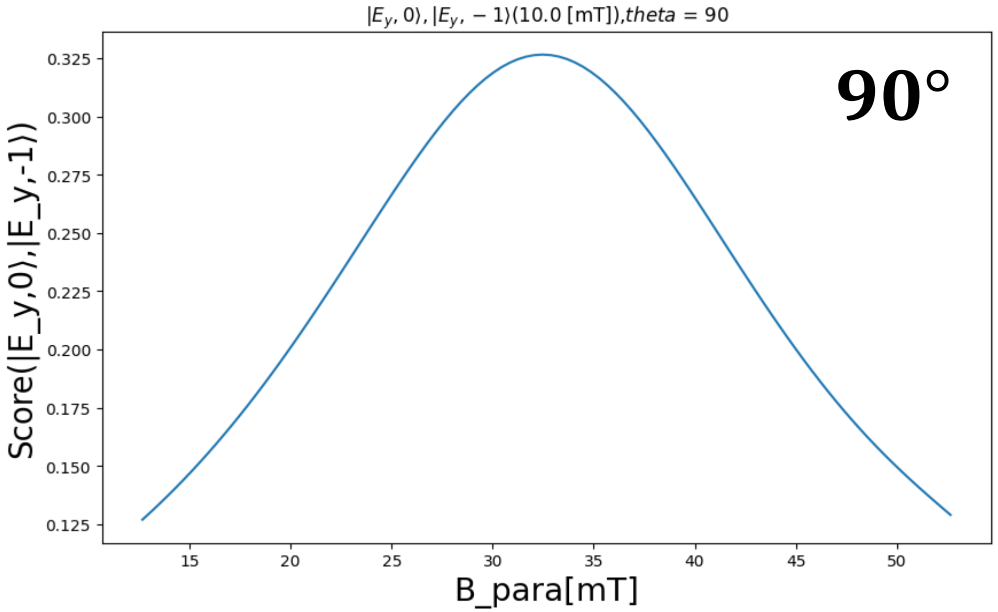
角度依存性



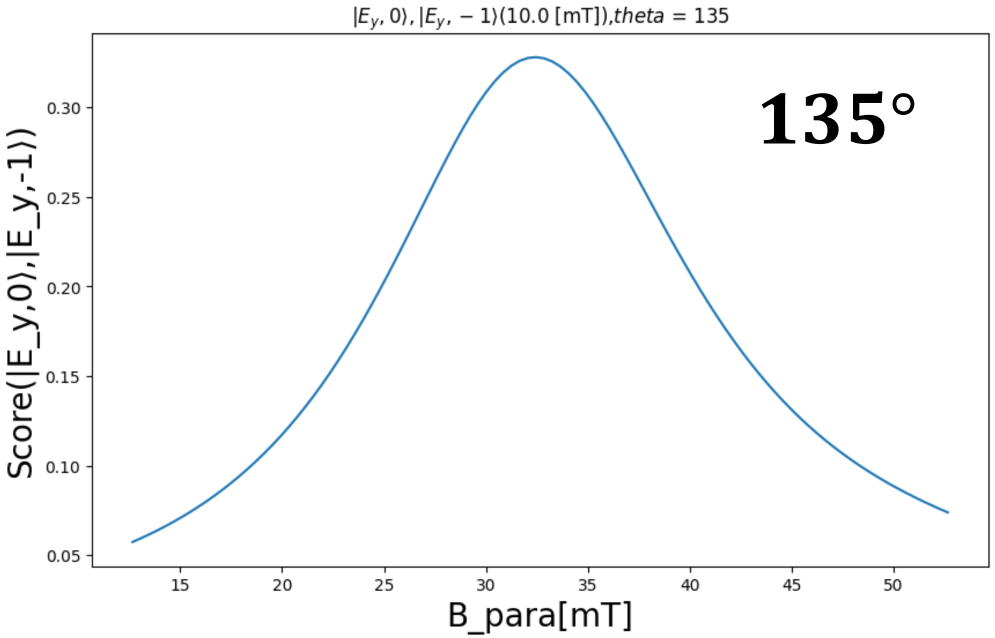
$|E_y,0\rangle,|E_y,-1\rangle$(10.0 [mT]),$theta$ = 0
0°
Score(|E_y,0⟩,|E_y,-1⟩)
B_para[mT]



$|E_y,0\rangle,|E_y,-1\rangle$(10.0 [mT]),$theta$ = 45
45°
Score(|E_y,0⟩,|E_y,-1⟩)
B_para[mT]



$|E_y,0\rangle,|E_y,-1\rangle$(10.0 [mT]),$theta$ = 90
90°
Score(|E_y,0⟩,|E_y,-1⟩)
B_para[mT]



$|E_y,0\rangle,|E_y,-1\rangle$(10.0 [mT]),$theta$ = 135
135°
Score(|E_y,0⟩,|E_y,-1⟩)
B_para[mT]

ハミルトニアン

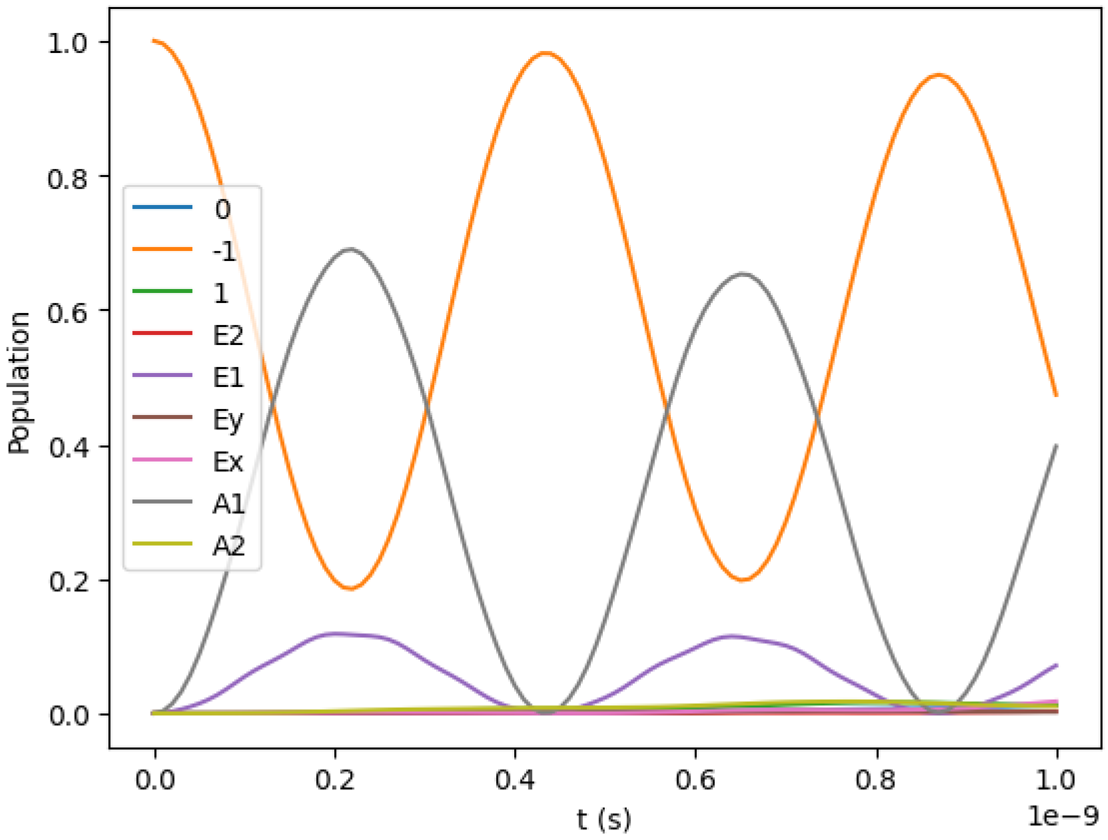
・B_x、B_yによる効果が効いてくるスピン基底の特定が可能
・B_perpの効果は角度θにほとんど依らず、その大きさで決まる

固有方程式

・B_perpにより、ESLACラインの線幅が変化し、また右にシフト

Sub issue2

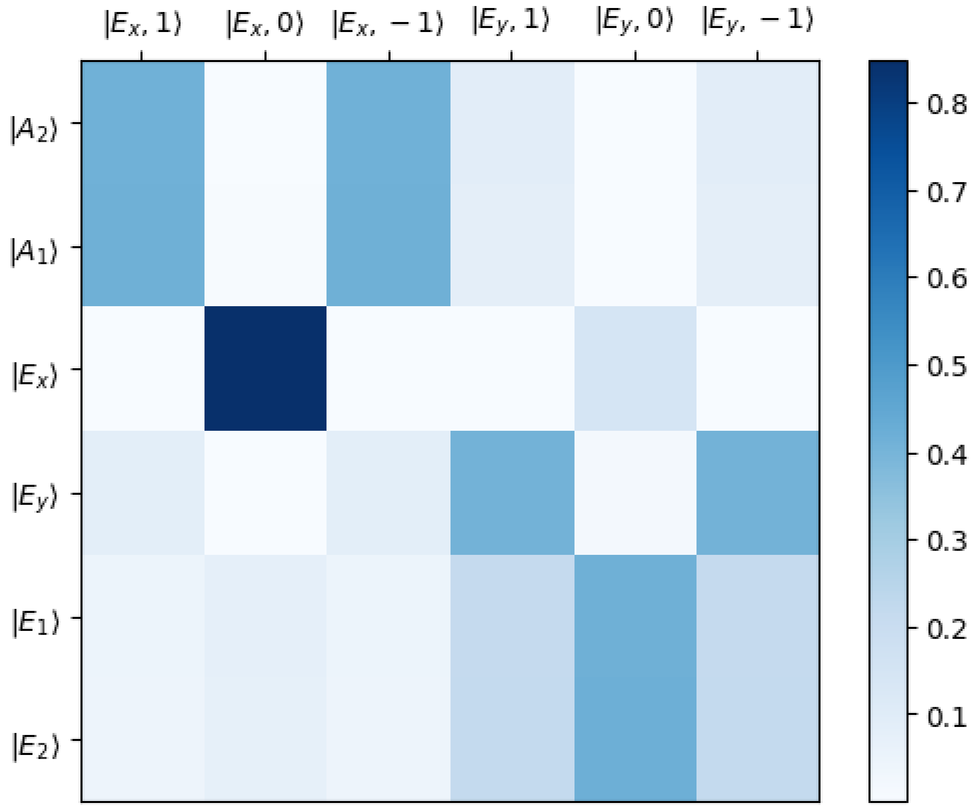
Rabi振動の確認

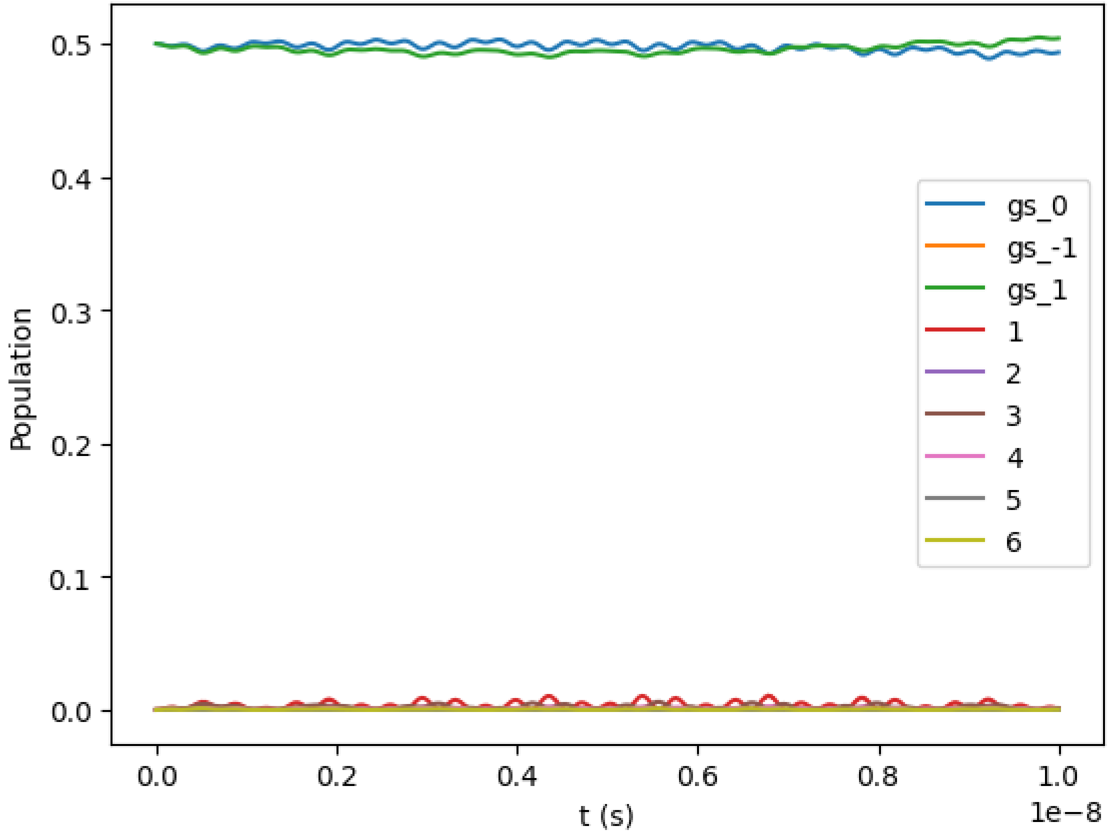


$(\Omega_0 = \Omega_1 = 1\ GHz)$

Sub issue2

Populationの計算(Acosta 2013 $B\_z = 0, \delta_{perp} = 10GHz \pm 15GHz$)

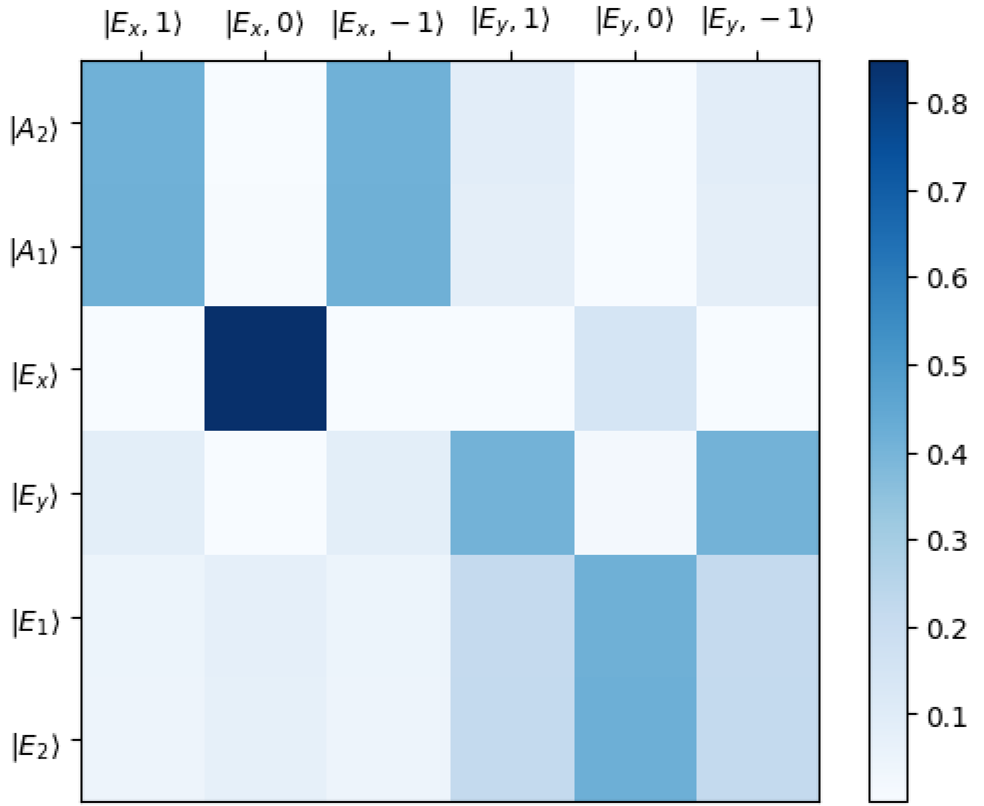


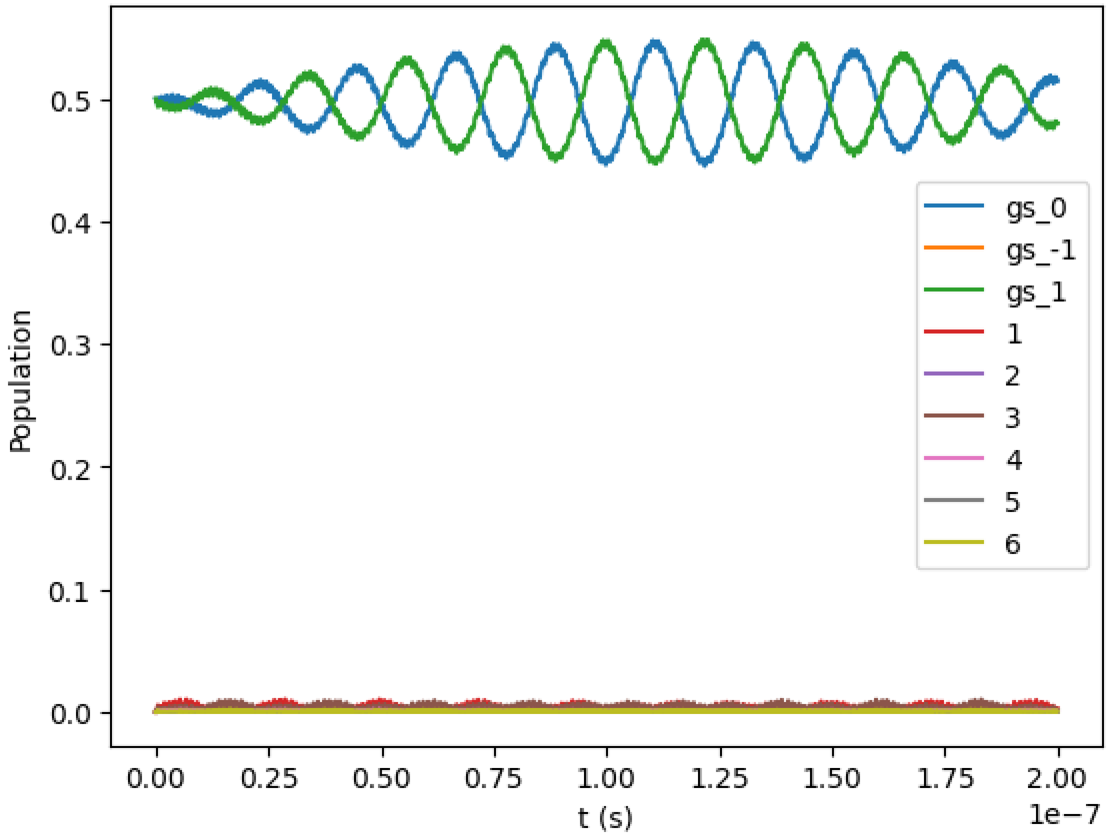


$$|D\rangle = \frac{1}{\sqrt{\Omega_0^2+\Omega_1^2}}(\Omega_1|0\rangle - \Omega_0|1\rangle), \qquad (\Omega_0 = \Omega_1 = 1\ GHz)$$

Sub issue2

Populationの計算(Acosta 2013 $B\_z = 0, \delta_{perp} = 10GHz \pm 15GHz$)

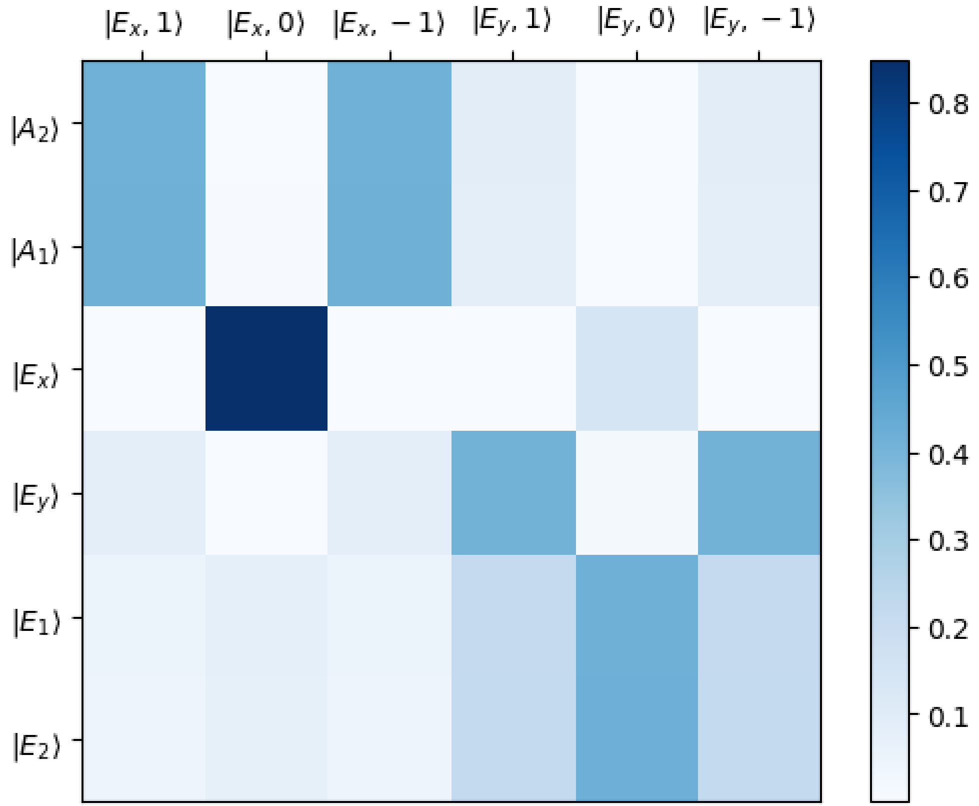


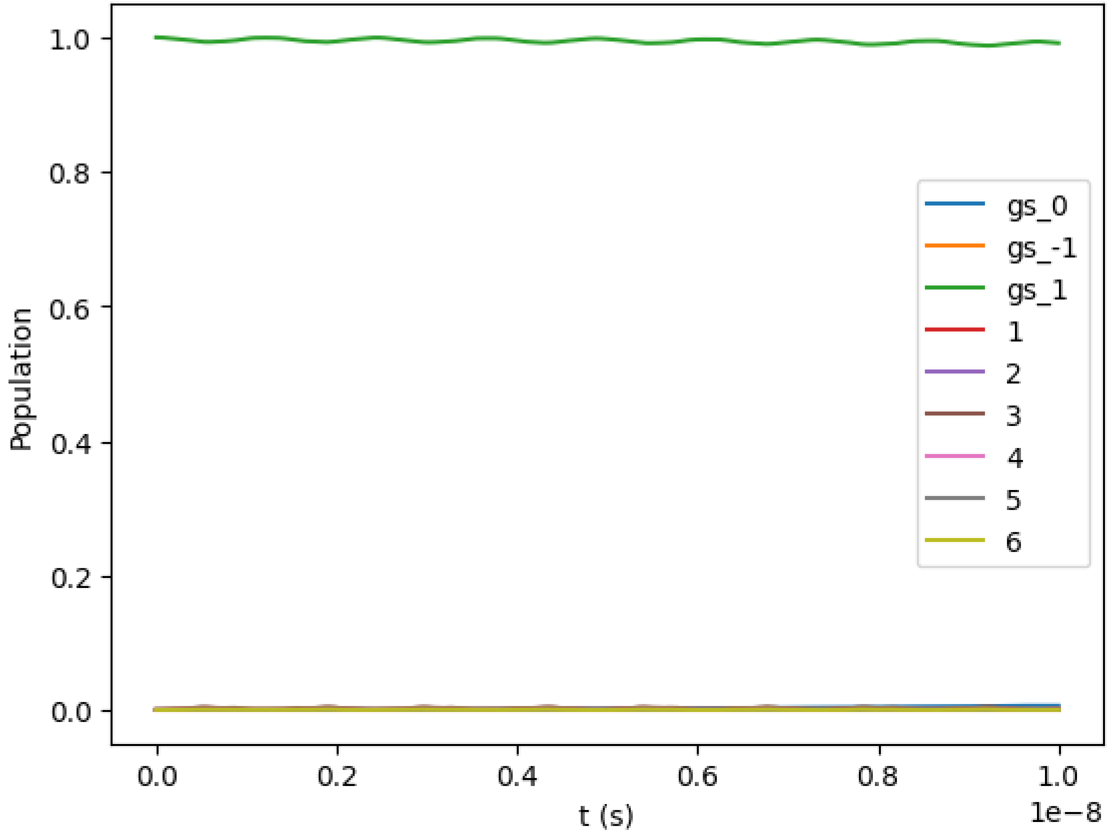


$$|D\rangle = \frac{1}{\sqrt{\Omega_0^2+\Omega_1^2}}(\Omega_1|0\rangle - \Omega_0|1\rangle), \qquad (\Omega_0 = \Omega_1 = 1\ GHz)$$

Sub issue2

Populationの計算(Acosta 2013 $B\_z = 0, \delta_{perp} = 10GHz \pm 15GHz$)



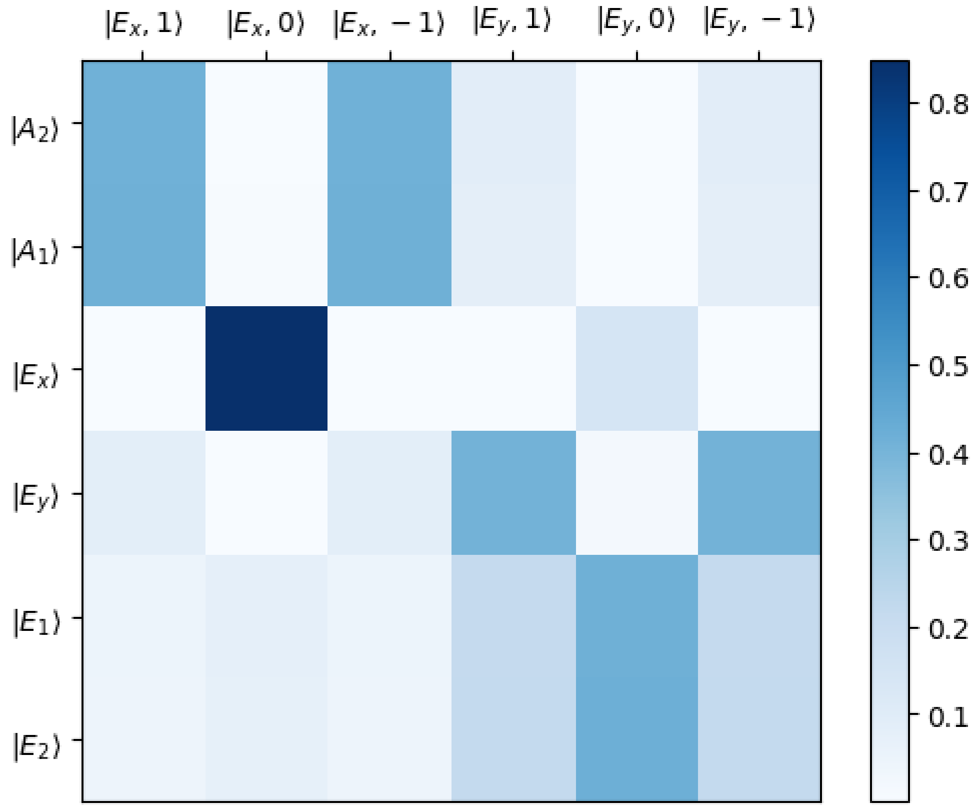


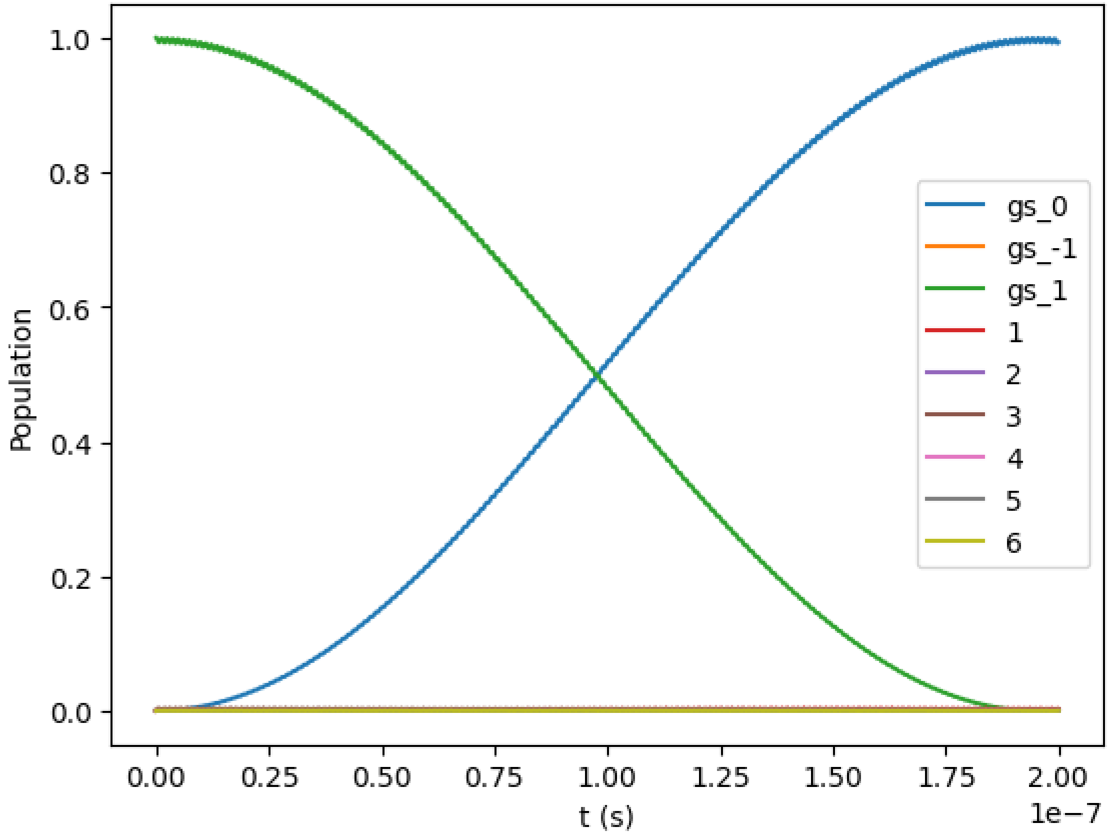
$$|D\rangle = \frac{1}{\sqrt{\Omega_0^2+\Omega_1^2}}(\Omega_1|0\rangle - \Omega_0|1\rangle),$$

$(\Omega_0 \gg \Omega_1 \quad \Omega_0 = 1GHz, \Omega_1 = 1MHz)$

Sub issue2

Populationの計算(Acosta 2013 $B\_z = 0, \delta_{perp} = 10GHz \pm 15GHz$)



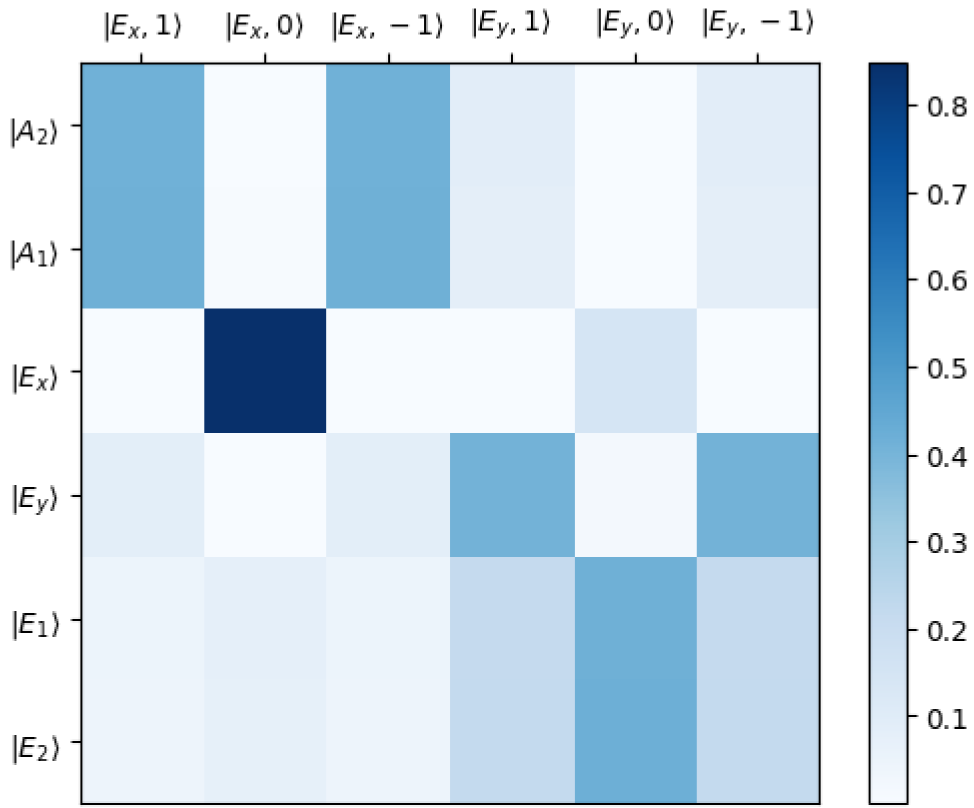


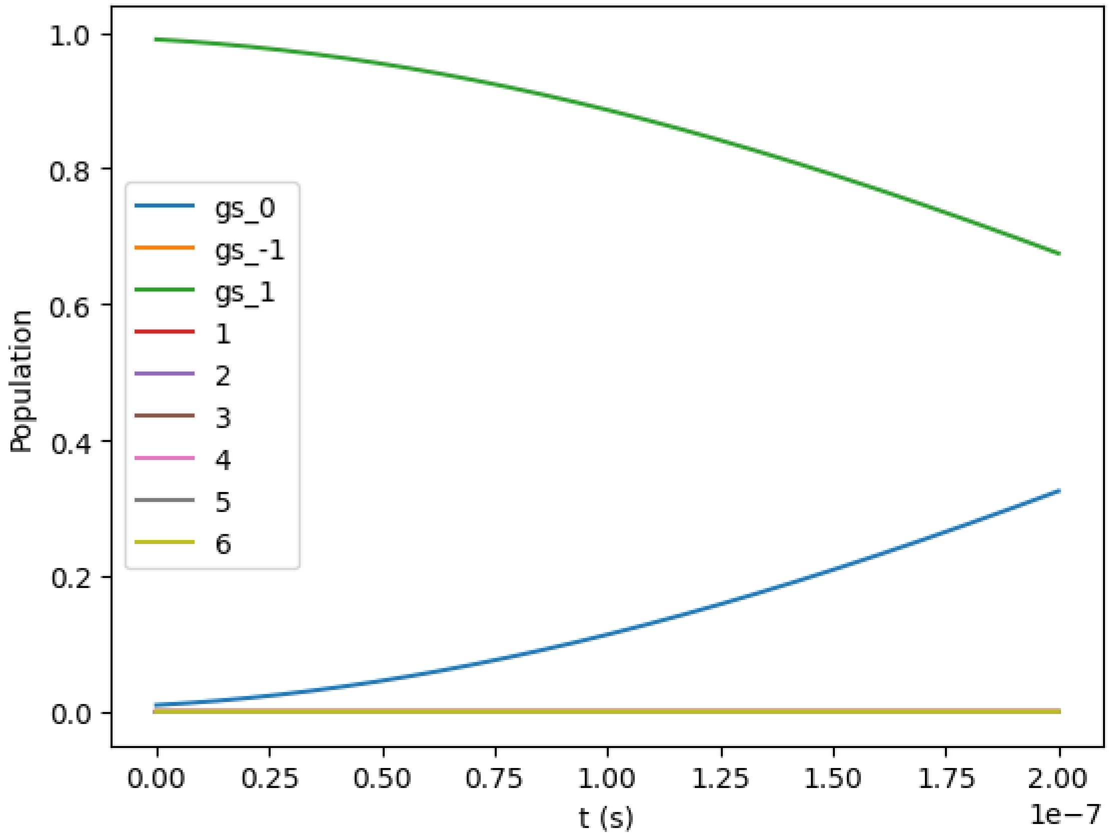
$$|D\rangle = \frac{1}{\sqrt{\Omega_0^2+\Omega_1^2}}(\Omega_1|0\rangle - \Omega_0|1\rangle),$$

$(\Omega_0 \gg \Omega_1 \quad \Omega_0 = 1GHz, \Omega_1 = 1MHz)$

Sub issue2

Populationの計算(Acosta 2013 $B\_z = 0, \delta_{perp} = 10GHz \pm 15GHz$)





$$|D\rangle = \frac{1}{\sqrt{\Omega_0^2 + \Omega_1^2}}(\Omega_1|0\rangle - \Omega_0|1\rangle),$$

$(\Omega_0 = 10MHz, \Omega_1 = 1MHz)$

$$|D\rangle = \frac{1}{\sqrt{\Omega_0^2+\Omega_1^2}}(\Omega_1|0\rangle - \Omega_0|1\rangle),$$

[34]. The fit parameters include a Rabi frequency conversion factor, $P_{\text{sat}} \equiv \pi \Delta\nu_{\text{nat}}^2 P_{\text{red}}/\Omega_0^2 = 2.4 \pm 1.1$ mW, the ratio $\Omega_1/\Omega_0 = 0.08 \pm 0.02$, which reflects the ensemble-averaged asymmetry in Λ transition strengths [38], and nuclear-spin-dependent ground-

From Acosta 2013